MILITARY SPECIFICATION

ANTENNA GROUP, OE-207/BR

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers the performance, design, manufacture, test, and acceptance requirements for the Antenna Group, OE-207/BR, hereinafter referred to as the antenna group.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17	Cables, Radio Frequency, Flexible And Semirigid, General Specification For
MIL-S-901	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication, And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-F-19207	Fuseholders, Extractor Post Type, Blown Fuse Indicating And Nonindicating, General Specification For
MIL-E-22200	Electrodes, Welding, Covered; General Specification For
MIL-C-23020	Cable, Coaxial (For Submarine Use)
MIL-C-24217	Connectors, Electrical, Deep Submergence, Submarine
MIL-C-24231	Connector, Plugs, Receptacles, Adapters, Hull Inserts, And Hull Insert Plugs, Pressure Proof, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, And In Specifications, Standards, And Technical Documents
MIL-STD-22	Welded Joint Design
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-248	Welding And Brazing Procedure And Performance Qualification
MIL-STD-271	Nondestructive Testing Requirements For Metals
MIL-STD-278	Fabrication Welding And Inspection; And Casting Inspection And Repair For Machinery, Piping And Pressure Vessels In Ships Of The US Navy
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-781	Reliability Design Qualification And Production Acceptance Tests: Exponential Distribution
MIL-STD-965	Parts Control Program
DoD-STD-1399, Section 300	Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)
MIL-STD-1472	Human Engineering Design Criteria For Military Systems, Equipment And Facilities

2.1.2 Other Government drawings and publications. The following other Government drawings and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

28672 A,B,C	Microfilm

NAVAL SHIP SYSTEMS COMMAND (NAVSHIP)

2477254	Fiberglass Mast Specifications, Submarine Antenna
4398613	Painting And Refinishing Fiberglass Masts And Radomes
4491190	Multifunction Mast Antenna Pre-installation Planning Data For TRIDENT

NAVAL ELECTRONIC SYSTEMS COMMAND (NAVELEX)

0104268	Electrical Interconnections Coordination Drawing
0104301	TRIDENT Multifunction Mast Antenna System (OE-207/BR) Coordination Drawing
0104321	TRIDENT Modification Mast Antenna Control Unit Outline Dimensions/Mounting

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PUBLICATIONS

NAVSEA

0900-LP-023-8070	Procedures For Painting Fiberglass Radomes And Camouflage Painting Of Submarine Fiberglass Faired Masts
0900-LP-074-4010	TRIDENT CCS Specification

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Electrical Code NFPA 70-1984

(Application for copies should be addressed to the National Fire Protection Association; 470 Atlantic Avenue, Boston, MA 02210.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3 REQUIREMENTS

3.1 General. The antenna group shall conform to the requirements of MIL-E-16400, except as specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.1.2 Antenna group description. The antenna group shall consist of the units specified in a through e (as specified by NAVSEA Drawing (DWG) 28672).

a. Radome and Antenna (Unit 1), AS-3076/BR (see 3.3.1)
b. Antenna Control Unit (ACU) (Unit 2), C-9724/BR (see 3.3.8)
c. Control and Radio Frequency (RF) Cable Assembly (Unit 3), CX-13001/BR (see 3.3.9)
d. Junction Box (Unit 4), J-3280/BR (see 3.3.10)
e. Hull Penetration Insert (Unit 5), MX-9651/BR (see 3.3.11)

3.2 General characteristics. The characteristics of MIL-E-16400 specified in a through d do not apply:

a. Maximum height
b. Surface ship installation
c. Shore installation
d. Standard hardware program (modular construction)

3.2.1 Environmental performance. The environmental service conditions of MIL-E-16400 specified in a through e do not apply:

a. Sunshine
b. Fungus
c. Wind velocity
d. Icing
e. Magnetic field

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3.2.1.1 Temperature. The antenna group shall conform to the temperature requirements of MIL-E-16400, except as specified in a and b:

a. Range 4 - For Units 2 and 4
b. Range 2 - For Units 1, 3, and 5

3.2.1.2 Shock. The antenna group shall withstand the Grade A, Class 1, Type A test for medium weight equipment in accordance with MIL-S-901 as modified by the APPENDIX.

3.2.1.3 Vibration. The antenna group shall withstand the Type I testing requirements of MIL-STD-167-1.

3.2.1.4 Inclination. The 60 degree inclination requirement of MIL-E-16400 shall apply to the ACU and junction box.

3.2.1.5 Wave slap. Outboard components of the antenna group (when the antenna is fully erected with associated mast) shall withstand a 0.25 kilograms per square centimeter (kg/cm^2) (500 pounds per square foot (lb/ft^2)) maximum projected area loading with a safety factor of 2.

3.2.1.6 Underwater speed. Outboard components subjected to hydrodynamic water flow shall be capable of withstanding a relative velocity of 12 knots (at submarine snorkel depth when the antenna is fully erected) without degradation of performance. Conditions assume a severe ship turning maneuver verging on hydrodynamic stall of the hydrofoil involved.

3.2.1.7 Airborne and structureborne noise. The antenna group shall conform to the airborne noise acceptance levels as specified in TABLE I. The antenna group shall conform to the structureborne noise acceptance levels specified in TABLE I-10 of NAVSEA 0900-LP-074-4010. Noise measurements and requirements shall be in accordance with APPENDIX 60.2 of NAVSEA 0900-LP-074-4010.

TABLE I. Airborne noise acceptance levels (decibels (dB)).

Unit	Frequency Range (Hertz (Hz))							
	37.5-75	75-150	150-300	300-600	600-1.2k 1/	1.2k-2.4k	2.4k-4.8k	4.8k-9.6k
1	90	85	80	80	75	75	75	75
2	72	68	64	numerical average between 300-4800				44
4	72	68	64	shall not exceed 45 dB				44

1/ Thousand

3.2.1.8 Electromagnetic pulse (EMP) protection. The antenna group shall be hardened or shall be intrinsically hard against the EMP resulting from an exo-atmospheric nuclear burst to survive the levels specified in 3.2.1.8.1 and 3.2.1.8.2.

3.2.1.8.1 External equipment. Equipment operated external to the hull shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times indicated in the individual equipment or subsystem specification after being subjected to a test signal having the waveform and amplitude specified in FIGURE 1.

3.2.1.8.2 Inboard equipment. All equipment and subsystem interface pins and terminals of leads, including grounds and neutrals, shall not exhibit any permanent malfunction, degradation of performance, or deviation from specified indications beyond the tolerances and recovery times indicated in the individual equipment or subsystem specification after being subjected to a test signal having either the waveform and common mode current level specified in FIGURE 2, whichever occurs first at the specific test frequency.

3.2.2 Human engineering. Human engineering shall be in accordance with 3.2.2.1 and 3.2.2.2.

3.2.2.1 Human engineering requirements. The antenna group shall conform to the human factors operational and maintenance design criteria of MIL-STD-1472 and the requirements specified in 3.2.2.2.

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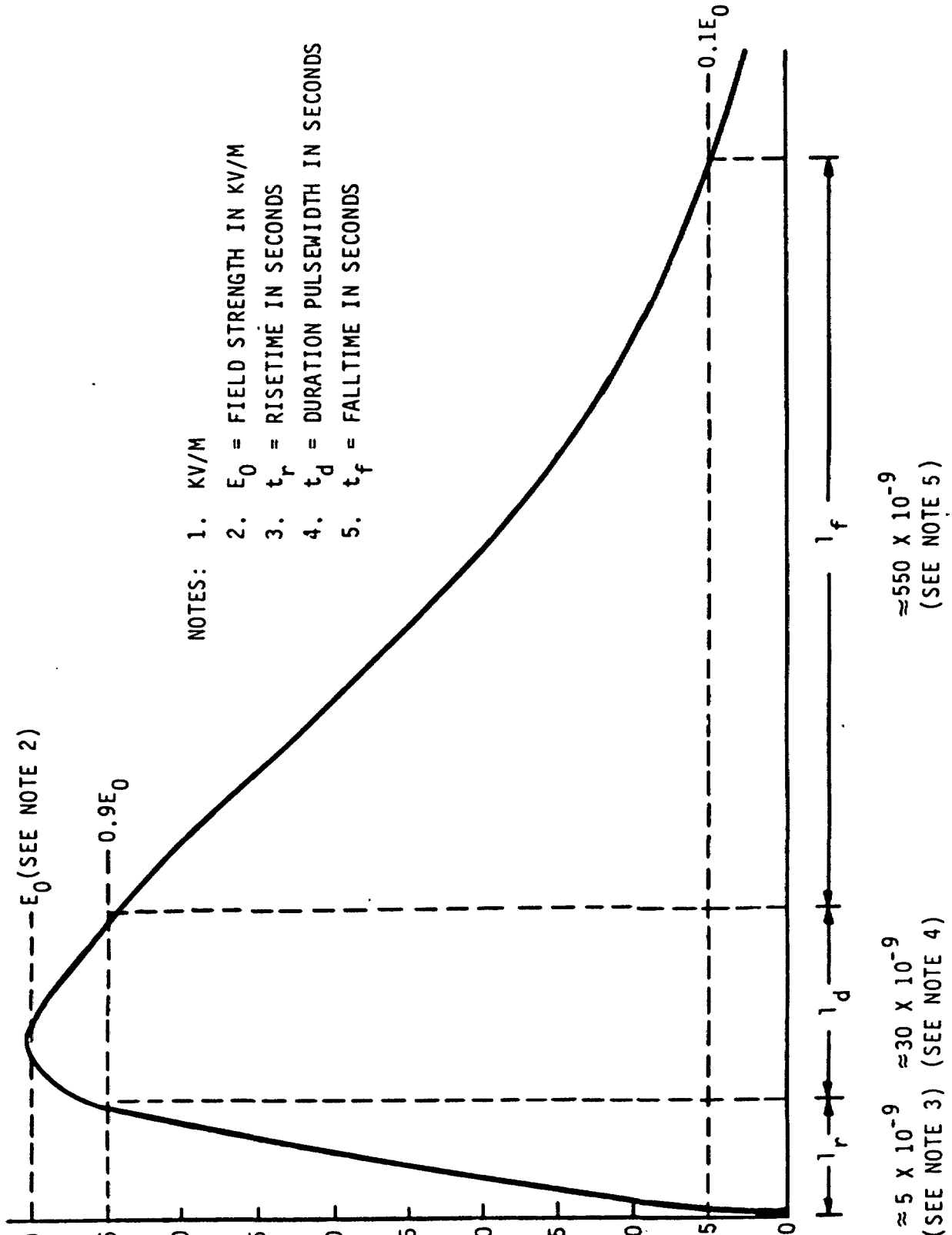
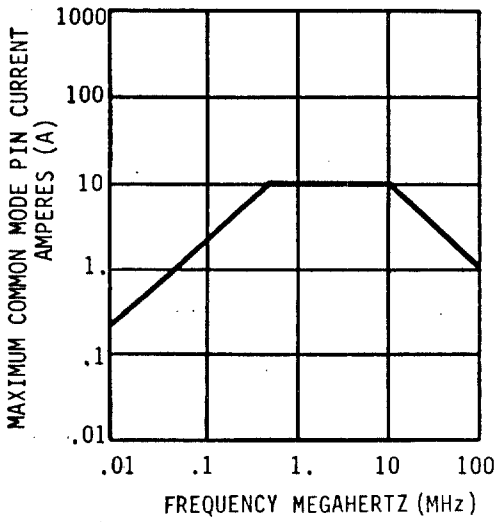
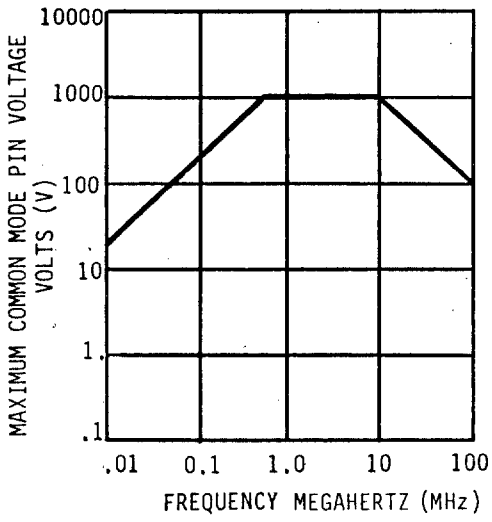


FIGURE 1. Test signal waveform.

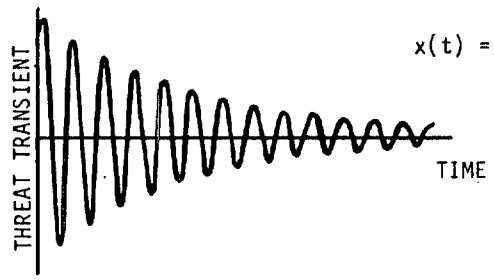
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INTERFACE PIN CURRENT FOR EQUIPMENT



INTERFACE PIN VOLTAGE FOR EQUIPMENT



$$x(t) = X(f_o) \text{ Sin } (2\pi f_o t)\ e^{(-\pi f_o t/Q)}$$

$$Q = 16 \pm 4$$

FIGURE 2. Test signal waveform characteristics.

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3.2.2.2 Controls, indicators, and panel layouts. The design of operator and maintainer panel layouts, controls, and consoles shall be in accordance with MIL-STD-1472. The abbreviations of MIL-STD-12 shall be used for labeling where applicable.

3.2.3 Parts, materials, and processes. The parts, materials, and processes shall be in accordance with the Parts, materials and processes paragraph of MIL-E-16400, except as specified in 3.2.3.1 through 3.2.3.3.

3.2.3.1 Parts control. Parts control shall be in accordance with MIL-STD-965, Procedure I.

3.2.3.2 Materials. The materials for the antenna group shall be as specified in 3.2.3.2.1 through 3.2.3.3.

3.2.3.2.1 Mercury and radioactive material. Parts used in the antenna group shall not contain, nor have come in direct contact with, mercury, mercuric compounds, or mercury-bearing instruments on devices employing only a single boundary containment during manufacture or inspection. No radioactive paints or coatings shall be used.

3.2.3.2.2 SUBSAFE requirements. All materials used in construction of the hull fitting insert and connector shall conform to SUBSAFE requirements and are subject to approval by NAVELEX (see 4.6).

3.2.3.2.2.1 SUBSAFE requirements (hull penetration). Portions of the antenna (hull penetration) to be supplied under this specification is in a SUBSAFE application aboard Navy submarines. SUBSAFE is a term used to identify certain materials in which maximum confidence is required. The materials utilized shall be manufactured under strict quality control procedures to assure strict compliance with all invoked specifications.

3.2.3.2.3 Lightweight material. Aluminum or other lightweight material (approved by NAVELEX) may be used in those units of the antenna group where the weight of the units can be significantly reduced without loss of strength, machine accuracy, or resistance to wear and corrosion. Use of magnesium is not permitted.

3.2.3.2.4 Cable. Special interconnecting cables shall require approval by NAVELEX prior to use and shall be furnished by the contractor as part of antenna group.

3.2.3.3 Painting. Painting shall be in accordance with MIL-E-16400, except that radome finishing shall be in accordance with NAVSHIPS DWG 4398613 and NAVSEA 0900-LP-023-8070.

3.2.4 Electrical characteristics. The electrical characteristics of MIL-E-16400 specified in a through c do not apply:

a. Dielectric withstanding voltage
b. Insulation resistance
c. Leakage current, phase to ground

3.2.4.1 Primary power. The antenna group shall operate from single-phase, 115 V, 60 Hz power having characteristics as specified for DoD-STD-1399, Section 300, Type I. Maximum power consumption shall be 525 watts (W).

3.2.4.2 Insertion loss. Insertion loss from the antenna base connector to the connector nearest each individual antenna (including the relay package, associated cable, diplexer, switch, and connector) shall not exceed 1.5 dB between 150 MHz and 400 MHz.

3.2.4.3 Electromagnetic compatibility (EMC). The antenna group shall be designed to operate compatibly in a submarine electromagnetic environment. At a mimimum, the antenna group shall conform to the requirements of MIL-STD-461 for Class A5 equipment. The required tests are CE01, CE03, CS01, CS02, CS06, RE01, RE02, RS01, RS02, and RS03. The limits and susceptibility levels shall be described in the approved EMC test plan.

3.2.4.4 Blown fuse indicators. Blown fuse indicators shall be provided for all fuses and shall be of the neon type using clear lenses, except where neon type indicators cannot be utilized due to the voltage involved. Where neon type indicators cannot be used, MIL-F-19207 shall apply.

3.2.5 Mechanical characteristics. The mechanical requirements of MIL-E-16400 specified in a through f do not apply:

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a. Gears and cams life expectancy
b. Dials and pointers for units having self-contained red illumination
c. Dials and pointers for units not having self-contained illumination
d. Windows
e. Time totalizing meters
f. Thermal sensor location

3.2.5.1 Construction. Construction of the antenna group shall be designed to permit rapid assembly and disassembly for ease of maintenance.

3.2.5.2 Enclosures. Except for interior surfaces with openings for cooling, the ACU and junction box shall have enclosures designed in accordance with the Enclosures paragraph of MIL-E-16400 and conforming to the dripproof (45 degrees) requirements of MIL-STD-108.

3.2.5.3 Magnetic permeability. The antenna group (except for the junction box) shall conform to the Magnetic materials paragraph of MIL-E-16400.

3.2.5.4 Controls. Controls with critical settings or which require setting only occasionally shall be provided with a locking device.

3.2.5.5 Indicator lamps. Indicator lamps shall conform to the Indicators and associated items paragraph of MIL-E-16400, except for dimmer controls.

3.2.6 Safety. The design of the antenna group shall conform to the Safety criteria paragraph of MIL-E-16400, except as specified in 3.2.6.1 through 3.2.6.7.

3.2.6.1 Equipment power connectors. Equipment power connectors for 115 volts alternating current (VAC) input power, and interconnecting alternating current (AC) power cable connectors between equipment, shall have only three pins designated A, B, and C. Pin assignment and color code shall be as shown in TABLE II.

TABLE II. Pin assignment and color code.

Pin designation	Conductor assignment	Color code
A	115 VAC return	White
B	Safety ground	Green
C	115 VAC high (hot)	Black

The black-and-white color code shall be maintained from the input connector pins to all components having the same voltage and frequency as the input power.

3.2.6.2 Equipment electrical power on-off switch. A switch for disconnecting equipment from all electrical power systems shall be mounted on the equipment front panel. The function of the switch shall be clearly labeled. The switch shall break all power conductors of the power circuit.

3.2.6.3 Power indicator lamp. A green power indicator lamp shall be mounted near the power on-off switch to indicate when the equipment is energized. The lamp shall be connected across the input power conductors on the load side of the switch.

3.2.6.4 Overcurrent device location. A fuse or an overcurrent trip unit of a circuit breaker shall be connected in series with each ungrounded conductor as determined by NFPA 70-1984 for grounded source power.

3.2.6.5 Material flammability. Material used as part of the antenna group shall conform to MIL-STD-454, Requirement 3.

3.2.6.6 Weight marking. Assemblies and subassemblies weighing in excess of 15.8 kilograms (kg) (35 pounds (lbs)) shall have the weight clearly marked on the external surface of the unit. The marking shall be as permanent as the normal life expectancy of the equipment.

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3.2.6.7 Power supply protection. Fault conditions ranging from open circuits to short circuits shall not cause damage to the power supply.

3.3 Detailed requirements. Detailed requirements shall be in accordance with 3.3.1 through 3.3.12.5.

3.3.1 Radome and antenna unit (Unit 1). The basic design concept of the radome and antenna unit is that of an epoxy-bonded fiberglass shell which houses and seals within it the antenna sections that perform the functions specified in a through f:

a.	Medium frequency (MF)/ high frequency (HF) transceive	2 MHz to 30 MHz
b.	Ultra high frequency (UHF)/ transceive line-of-sight (LOS)	225 MHz to 400 MHz
c.	Very low frequency (VLF)/ low frequency (LF) receive	10 kHz to 160 kHz
d.	Navy Navigation Satellite System (NAVSAT) receive	149.988 MHz and 399.963 MHz
e.	UHF satellite communications (SATCOM) transceive	240 MHz to 320 MHz
f.	Identification, friend or foe (IFF) transceive	950 MHz to 1150 MHz

The antenna sections shall operate individually or in any combination of a, b, c, f; a, c, d, f; or a, c, e, f. VLF reception may be impaired by MF/HF transmissions.

3.3.1.1 Mechanical requirements. The radome and antenna unit shall be capable of mounting and operating within a retractable faired mast in the sail of a TRIDENT submarine conforming to NAVSHIPS DWG 4491190. The radome cross section and base header shall conform to NAVSHIPS DWG 2477254.

3.3.1.2 Dimensions. The maximum physical dimensions of the radome and antenna unit shall not exceed 624.8 centimeters (cm) (246 inches (in.)) in length and a cross section of 20.3 cm by 31.75 cm (8 in. by 12.5 in.). Provision shall be made for lifting sling attachment points.

3.3.1.3 Hydrostatic pressure requirements. Each antenna unit shall be capable of withstanding a maximum hydrostatic pressure of 70.3 kg/cm^2 (1000 pounds per square inch (psi)) without structural failure, water absorption in the radome walls, or leakage beyond the indicated seals. The primary seal is defined as that formed by the crush-ring seal and external plug. The secondary seal comprises the O-ring and the internal plug.

3.3.1.4 Weight. The weight of the radome and antenna unit completely assembled shall not exceed 408.2 kg (900 lbs.).

3.3.2 MF/HF Antenna (Unit 1A1). The MF/HF antenna proper shall consist of a helical conductor wound within a pregrooved plastic form, on which will be mounted a metallic capacitance top load. Inside the helix and top load shall be a shorting device comprised of a metallic shorting tube with shorting brushes at each end to complete the circuit between the helix and top load.

3.3.2.1 MF/HF antenna performance characteristics. The MF/HF antenna shall have the performance characteristics specified in a through f:

a. Frequency range. The range of operating frequencies shall extend from 2.0 MHz to 30.0 MHz at power levels up to 1 kilowatt (kW).

b. Impedance. The input impedance shall be 50 ohms nominal.

c. Voltage standing wave ratio (VSWR). The tuned VSWR shall be 2:1 or less over the frequency range of 2.0 MHz to 30.0 MHz.

d. Pattern. The azimuth radiation pattern shall be omnidirectional $\pm$1 dB.

e. Power input. The average power input capability shall be 1 kW continuous.

f. Bandwidth. The 3 dB bandwidth of the antenna between 2.0 MHz and 2.5 MHz shall be 15 kHz or greater; between 2.5 MHz and 30 MHz the bandwidth shall be 25 kHz or greater.

3.3.2.2 Efficiency. The antenna efficiency (in HF only mode) with the top and base of the radome 668 cm (263 in.) and 43.2 cm (17 in.), respectively, above the snorkel waterline shall be as specified in a through c:

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a.	2.0 MHz to 2.49 MHz	5 percent or greater
b.	2.5 MHz to 4.0 MHz	15 percent or greater
c.	4.01 MHz to 30 MHz	50 percent or greater

3.3.3 VLF/LF antenna (Unit 1A7). The VLF/LF antenna shall consist of separate loops that are installed for maximum reception when the direction of the incoming signal is either parallel or perpendicular to the forward and after axis of the radome and antenna unit (Unit 1).

3.3.3.1 VLF/LF antenna characteristics. The VLF/LF antenna shall have the performance characteristics specified in a and b:

a. Frequency range. The range of operating frequencies shall extend from 10 kHz to 160 kHz.

b. Effective height. The effective height of each loop shall be at least 0.7 millimeters (mm) at 20 kHz and 4.2 mm at 120 kHz.

3.3.3.1.1 Antenna pattern. The VLF/LF antenna shall have two bi-directional (figure-eight) patterns displaced 90 degrees ±0.5 degree in the horizontal plane across the frequency range of 10 kHz to 160 kHz. Throughout this frequency range, the four maximum values of these orthogonal response patterns shall be approximately equal, with differences not to exceed 2 dB. When measured at 20 kHz, decoupling between the loops shall be at least 34 dB.

3.3.3.1.2 Loop Q. The in-air Q of each loop inductor shall have a value greater than 135 at a frequency of 20 kHz measured in free space. The in-air Q of each loop inductor at the antenna radome base shall have a value greater than 65 kHz to 20 kHz.

3.3.3.1.3 Loop inductance. Each single plan loop inductor shall have a nominal inductance of 515 ±15 microHenries (μH) at a frequency of 20 kHz measured at the antenna radome base. The values of the two windings shall be within ±1 percent.

3.3.3.2 Depth transducer requirements. The VLF/LF antenna section shall have a means of providing a voltage output representing depth of the antenna below the surface.

3.3.3.2.1 Voltage output. The maximum output voltage from the depth transducer shall be 5 V ±0.25 V. The voltage shall vary from 0 V ±0.1 V to maximum representing 0 meter (m) to 30.48 m (100 feet (ft)). Scale factor for these values shall 0.164 volts per meter (V/m) (0.05 volts per feet (V/ft)).

3.3.3.2.2 Impedance. The output impedance of the depth transducer and associated circuitry shall be low enough to drive a 50 kohm load at the voltage output specified in 3.3.3.2.1.

3.3.3.2.3 Pressure. The depth transducer shall be capable of withstanding pressure up to 70.3 kg/cm^2 (1000 psi).

3.3.3.3 VLF/LF output requirements. The requirements specified in 3.3.3.3.1 through 3.3.3.3.7.2 include the loop antenna, cabling, connectors, and the ACU and are measured at the output of the junction box (Unit 4).

3.3.3.3.1 Preamplifiers. Preamplifiers shall be provided at the base of the antenna assembly for each VLF/LF loop such that the effective height of each loop at the junction box output when terminated in 50 ohms is 60 mm ±20 percent at 20 kHz. The preamplifiers shall be capable of being bypassed by removal of power supply voltage.

3.3.3.3.2 Frequency response. The output of each loop at the junction box output shall have a frequency response when referenced to 100 kHz specified in TABLE III:

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TABLE III. Frequency Response

First frequency	Second response
10 kHz to 20 kHz	+0 dB to -4 dB
20 kHz to 50 kHz	±2 dB
50 kHz to 70 kHz	±0.75 dB
70 kHz to 130 kHz	±0.25 dB
130 kHz to 150 kHz	±0.75 dB
150 kHz to 160 kHz	±2 dB

3.3.3.3.3 Phase difference. The maximum phase difference between loops shall be less than 20 nanoseconds (ns) at 100 kHz.

3.3.3.3.4 Phase shift variation. The maximum phase shift variation over a 100 hour interval shall be less than 45 milliseconds (ms). The root-mean-square (rms) phase shift variation over the same period shall be less than 15 ms. The variation shall be measured for the antenna, preamplifier, and associated cables and connectors within the antenna radome unit.

3.3.3.3.5 Harmonic distortion. Harmonic distortion at a maximum output voltage of 0.1 V shall not exceed 1 percent (with the output terminated in 50 ohms).

3.3.3.3.6 Phase characteristics. A least mean square fit line, drawn through a plot of phase shift versus frequency in linear coordinates over the frequency range of 70 kHz to 130 kHz shall intercept the phase shift axis at zero frequency within the range of ±25 degrees. The phase shift shall be measured from the electromagnetic field in air incident on the antenna to the output of the junction box. The phase reference for each loop shall be as shown on NAVELEX DWG 0104268. The maximum deviation of phase response from the least mean square fit line shall be ±1 degree from 70 kHz to 130 kHz and ±2 degrees from 50 kHz to 70 kHz and 130 kHz to 150 kHz.

3.3.3.3.7 Sensitivity. Sensitivity shall be in accordance with 3.3.3.3.7.1 and 3.3.3.3.7.2.

3.3.3.3.7.1 VLF sensitivity. A field intensity of 25 microvolts per meter (μV/m) at the loop center (single plane) shall provide a minimum signal-to-noise ratio (SNR) of 20 dB. Sensitivity measurements shall be made from a 500 ohm transmission line fed by a standard signal generator at a frequency of 20 kHz. The 500 ohm transmission line shall be a single wire as one conductor, and the shielded room the other conductor. The transmission line shall be terminated in its characteristic impedance with a non-inductive resistance. The bandwidth shall be normalized to 50 Hz. A simulator may be used if previously approved by NAVELEX.

3.3.3.3.7.2 LF sensitivity. The sensitivity of the loop antenna at 100 kHz shall be such that a field intensity of 50 μV/m shall produce a SNR greater than or equal to 1 in a bandwidth of 70 kHz.

3.3.4 VHF/UHF antenna (Unit 1A2). The VHF/UHF antenna shall be a biconical vertical dipole to be used for UHF LOS transmission and reception, NAVSAT reception, and UHF SATCOM low angle transmission and reception.

3.3.4.1 VHF/UHF antenna characteristics. The VHF/UHF antenna shall have the performance characteristics specified in a through g:

a. Frequency range. The VHF/UHF antenna shall operate over a frequency range of 225 MHz to 400 MHz plus 150 MHz.
b. Impedance. The input impedance shall be 50 ohms nominal.
c. VSWR. The VSWR shall be 2:1 or less over the frequency range of 225 MHz to 400 MHz.
d. Pattern. The antenna radiation pattern shall be omnidirectional ±2 dB in the horizontal plane. The vertical radiation pattern shall have a figure-eight shape representative of a vertical dipole. This pattern shall be maintained at all operating frequencies.
e. Power input. The antenna continuous wave power capability shall be 100 W over the frequency range of 225 MHz to 400 MHz.

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f. Gain. The antenna shall have a minimum gain of 1 dB with respect to a vertically polarized isotropic radiator from 225 MHz to 400 MHz and -1.5 dB at 150 MHz.

g. Polarization. The antenna shall be linear vertically polarized.

3.3.4.2 RF feed system losses. Neglecting losses in the unspecified length of interconnecting cable between the pressure hull fitting and the ACU input, overall losses of the RF path from the connector output of the antenna to the output of the antenna control unit shall not exceed 4 dB at 150 MHz and 400 MHz. The specified feed losses shall include losses in the antenna matching or isolation devices, filters, connectors, relays, couplers, and switches with the preamplifier bypassed.

3.3.4.3 UHF-LOS performance requirements. The VHF/UHF antenna will provide for transmission and reception of UHF-LOS continuous wave signals up to a power level of 100 W over the frequency range of 225 MHz to 400 MHz.

3.3.4.4 NAVSAT performance requirements. The antenna shall be compatible with the performance capabilities of the NAVSAT. The antenna shall meet the requirements specified in a and b and in 3.3.4.1:

a. Phase shift variation. The maximum phase shift variation for NAVSAT signals at the ACU output, during a 20 minute interval shall not exceed 1 degree.

b. Frequency. The antenna shall be capable of simultaneous reception of 149.988 MHz and 399.963 MHz.

3.3.4.5 SATCOM (low-angle) performance requirements. The VHF/UHF antenna shall be compatible with a SATCOM transceiver operating in a simplex mode over the frequency range of 240 MHz to 320 MHz.

3.3.5 IFF antenna (Unit 1A6). The IFF antenna shall be a modified conical antenna to be used for IFF transmission and reception.

3.3.5.1 IFF antenna characteristics. The IFF antenna shall have the performance characteristics specified in a through g:

a. Frequency range. The IFF antenna shall operate over the frequency range of 950 MHz to 1150 MHz.

b. Impedance. The input impedance shall be 50 ohms nominal.

c. VSWR. The VSWR of the IFF antenna shall be equal to or less than 2:1 over the frequency range of 950 MHz to 1150 MHz.

d. Pattern. The IFF antenna shall be omnidirectional $\pm$2 dB in the horizontal plane. The vertical pattern shall approximate a figure eight.

e. Power input. The IFF antenna shall handle at least 1 kW peak power with a 1 percent duty cycle.

f. Gain. The gain of the IFF antenna at the pattern maximum shall be greater than 1 dB with respect to a vertically polarized isotropic radiator.

g. Polarization. The IFF antenna shall have linear vertical polarization.

3.3.5.2 RF feed system losses. The maximum insertion loss for the IFF RF signal path from the antenna radiator connector to the junction box output shall be less than 7.5 dB at 1030 MHz and 1090 MHz.

3.3.6 SATCOM antenna (Unit 1A3). The SATCOM antenna shall be a quadrafilic helix for reception and transmission of SATCOM signals in an overhead mode. The SATCOM antenna shall be compatible with a SATCOM transceiver operating in a simplex mode.

3.3.6.1 SATCOM antenna characteristics. The SATCOM antenna shall have the performance characteristics specified in a through g:

a. Frequency range. The SATCOM antenna shall operate over the frequency range of 240 MHz to 320 MHz.

b. Impedance. The SATCOM antenna shall have a nominal impedance of 50 ohms.

c. VSWR. The VSWR of the SATCOM antenna shall not exceed 2:1, measured at the antenna base, over the frequency range of 240 MHz to 320 MHz.

d. Pattern. The SATCOM antenna shall have a pattern which is omnidirectional $\pm$1 dB in the horizontal plane at an elevation angle 30 degrees and unidirectional in the vertical plane with the maximum at 90 degrees.

e. Power input. The SATCOM antenna shall have an input power capability of 100 W over the frequency range of 240 MHz to 320 MHz.

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f. Gain. The SATCOM antenna shall have a minimum gain (at an elevation angle of 90 degrees) of 1 dB relative to a vertically polarized isotropic radiator.

g. Polarization. The SATCOM antenna shall have right-hand circular polarization.

3.3.6.2 RF feed system losses. Neglecting losses in the unspecified length of interconnecting cable between the pressure hull fitting and the ACU input and overall losses of the RF path from the connector output of the antenna to the output of the ACU shall not exceed 4 dB at 245 MHz and 320 MHz. The specified losses shall include losses in the antenna matching or isolation devices, filters, connectors, relays, couplers, and switches with the preamplifier bypassed.

3.3.7 VHF/UHF preamplifier. A low noise preamplifier shall be provided for reception of VHF/UHF signals from both the VHF/UHF antenna and the SATCOM antenna. The preamplifier shall be mounted within the radome and antenna unit as near as possible to the feed points of the two antennas. The maximum noise figure shall be 4 dB from 150 MHz to 400 MHz with a minimum gain of 25 dB. The 1 dB compression point of the preamplifier shall occur with a minimum of -10 decibels referred to one milliwatt (dBm) input. Protection from burnout by UHF transmissions from adjacent antennas shall be included. A fail-safe means for bypassing the preamplifier during the transmit mode shall be included.

3.3.8 ACU (Unit 2). The ACU shall be located in the radio room on TRIDENT submarines. The ACU shall house the power supply, reflectometer, and various control and indicating circuits necessary for remote operation of the antenna (Unit 1). The ACU shall also incorporate a capability to perform as a slave to the TRIDENT radio room control monitor and test console (CMT) for command functions.

3.3.8.1 Elapsed time meters. Elapsed time meters are not required.

3.3.8.2 Locking devices. Locking devices shall be in accordance with the Shaft locking devices paragraph of MIL-E-16400.

3.3.8.3 Physical dimensions. The ACU shall be compatible with the standard 48.26 cm (19 in.) Langley rack and have the following maximum dimensions: 44.45 cm (17.5 in.) wide by 31.115 cm (12.25 in.) high by 55.88 cm (22 in.) deep. NAVELEX DWGS 0104301 and 0104321 shall be used to specify the physical interface with the radio room.

3.3.8.4 Weight. The ACU shall weigh no more than 49.9 kg (110 lbs).

3.3.8.5 TRIDENT integrated radio room (IRR) interface. To ensure compatability with the IRR, the ACU shall conform to the criteria specified in a through e:

a. The ACU shall fit into a rack drawer having a locking provision, slides, and a tilting mechanism for maintenance and service from the front and shall be removeable by no more than two men. Rear panel access is not required.

b. All input and output cabling shall enter from the ACU rear and interface with the IRR cabinet.

c. Internal logic circuits shall be incorporated to allow control by the IRR CMT system.

d. Maximum power consumption shall not exceed 525 W as specified in 3.2.4.1. The average and peak heat loads shall not exceed 146.5 W (500 British thermal units per hour (BTU/hr)) and 205.1 W (700 BTU/hr), respectively.

e. Control panel presentation shall conform to human engineering procedures adapted for IRR.

3.3.9 Control and RF cable assembly (Unit 3). The control and RF cable assembly (Unit 3) shall be composed of an outboard cable with associated end fittings. The outboard cable assembly shall be in accordance with MIL-C-23020, MIL-C-24217, and MIL-C-24231 and shall conform to the cold bend requirements of MIL-C-17. The cable assembly shall complete the control RF circuitry between the antenna and the hull fitting receptacle. The cable assembly shall operate under adverse conditions of temperature and in the severe environment of the sea. The cable assembly shall withstand the high voltage and power requirements associated with the antenna.

3.3.9.1 Attenuation. The attenuation of the coaxial cable used in the outboard cable assembly (including connectors) shall be as specified in a and b:

a. At 400 MHz: 1.95 dB nominal, 2.2 dB maximum
b. At 1200 MHz: 3.6 dB nominal, 4.25 dB maximum

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3.3.9.2 Hydrostatic pressure requirements. The outboard wetted portion of each cable assembly shall withstand 70.3 kg/cm^2 (1000 psi) of external pressure for a period of 1 hour without evidence of leakage or degradation of the cable or fitting. The insulation resistance between each conductor and all other conductors in the cable and to ground, after exposure to pressure, shall not be reduced below 1000 megohms, when measured immediately after exposure to pressure.

3.3.9.2.1 Antenna connector and hull fitting connector. The connectors of the antenna and hull fitting inserts of the control and RF cable assembly shall conform to the hydrostatic requirements of MIL-C-24231.

3.3.9.3 Hydraulic shock. The connectors of the antenna and hull fitting insert of the control and RF cable assembly shall pass three blows, without failure, of 0.471 kilograms per second per square centimeter (kg-second/cm^2) (6.7 pounds per square inch (psi-second)) hydraulic shock superimposed on a static pressure of 66.1 kg/cm^2 (940 psi). The peak impulse pressure shall be 773.4 kg/cm^2 (11,000 psi) and shall decay exponentially to the static pressure.

3.3.10 Junction box (Unit 4). The junction box (Unit 4) shall be the junction point for interconnecting wiring between the hull fitting connector of the control and RF cable assembly (Unit 3) and the ACU (Unit 2). The junction box shall also provide the IFF output connector and the signal insertion points specified in 3.3.12.3.

3.3.11 Hull penetration insert (Unit 5). The hull penetration insert shall consist of an insert assembly with a receptacle on the outboard side and a cable assembly on the inboard side. The inboard cable assembly shall be provided with connectors required to complete the circuit between the hull receptacle and the inboard junction box. The hull penetration liner is not provided as part of this assembly.

3.3.11.1 Hydrostatic pressure requirements. The hull penetration insert shall conform to the hydrostatic pressure requirements of MIL-C-24231.

3.3.11.2 Underwater explosion requirements. The hull insert assembly shall pass underwater explosion tests in accordance with the APPENDIX.

3.3.11.3 Hydraulic shock. The hull penetration insert shall pass three blows, without failure, of 0.471 kg-second/cm^2 (6.7 psi-second) hydraulic shock superimposed on a static pressure of 66.1 kg/cm^2 (940 psi). The peak impulse pressure shall be 773.4 kg/cm^2 (11,000 psi) and shall decay exponentially to the static pressure.

3.3.12 CMT provisions. The antenna group shall incorporate provisions for remote control, monitoring, and testing of the antenna operation by the IRR CMT console. All signals, except for analog signals, shall use semiconductor differential line drivers and differential line receivers compatible with National Semiconductor types 7820 and 7830, or equal. Logic level 0 shall be the ON condition. The provisions in 3.3.12.1 through 3.3.12.5 shall be incorporated to allow interfacing with the IRR.

3.3.12.1 Remote control mode. The remote control mode, in which the CMT console of the IRR assumes control of the antenna operation, shall be selected from the ACU. In this mode, the ACU shall accept up to 20 control functions from the IRR to provide complete operation of the antenna group. The control functions in a through e shall be required:

a. HF tune up
b. HF tune down
c. HF tap up
d. HF tap down
e. HF ground No. 1
f. HF ground No. 2
g. SEL HF only
h. SEL UHF preamp OFF
i. SEL UHF NAVSAT
j. SEL UHF lo-angle
k. SEL UHF hi-angle
l. SEL HF/UHF test (VLF preamp OFF)

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3.3.12.2 Monitor lines. The ACU shall provide 20 monitor lines for continuous reporting of the antenna configuration and status to the IRR CMT console. The monitor lines shall be provided in both the local control operating mode and the remote control operating mode as as specified in a through j:

a. UHF NAVSAT
b. UHF lo-angle
c. UHF hi-angle
d. UHF preamp OFF
e. AC power ON
f. Remote control
g. HF ground No. 1
h. HF ground No. 2
i. HF only
j. HF/UHF test (VLF preamp OFF)

3.3.12.2.1 Analog function. The analog function shall be as specified in a through h:

a. HF tune position; +5 volts direct current (VDC) to +45 VDC, accuracy 1 part in 800
b. HF tap position; 0 VDC to +45 VDC, accuracy 1 part in 800
c. Pressure transducer; 0 VDC to 5 VDC
d. HF VSWR; 0 VDC to 5 VDC
e. Communications Mode Switch, AN/WSC-3 (to antenna); closed during SATCOM mode on, open during LOS mode on
f. SATCOM keyline AN/WSC-3 (to antenna); +12 VDC during SATCOM transmit
g. UHF transmitter interlock (from antenna); switch closed, except during NAVSAT reception
h. HF transmit interlock (from antenna); switch closed, except during mode switch operation

3.3.12.3 Signal insertion points. Provision shall be made for the antenna group to accept RF test signals to determine proper operation of MF/HF and VHF/UHF inboard cables from the junction box to the IRR and the VLF/LF, MF/HF, and VHF/UHF antennas. The VSWR of all test inputs shall be less than 2:1 on a nominal 50 ohm transmission line. Test provisions shall be as shown in TABLE IV.

3.3.12.3.1 Minimum coupling level. The minimum coupling level refers to the insertion loss from injection port to receive port. This level from each insertion point to the corresponding antenna system (without preamplifiers in the signal path) shall be equal to, or greater than, the values in TABLE IV.

3.3.12.4 Cable and connector interface. Transmission lines, control cables, connectors, and pin assignments shall be in accordance with NAVELEX DWGs 0104268 and 0104301.

3.3.12.5 Electrical interface. Impedance, signal, and control-voltage levels shall be in accordance with NAVELEX DWG 0104268.

3.4 Reliability. The lower mean-time-between-failure (MTBF) of the antenna group (θ_1 as defined in MIL-STD-781) shall be 850 hours.

3.5 Maintainability. The antenna group inboard components shall be a mean-corrective-maintenance-time (M_{ct}) not exceeding 10 minutes and a maximum-corrective-maintenance-time (M_{maxct}) (95th percentile) not exceeding 1 hour when corrective maintenance is accomplished at the organizational level by replacement of lowest assemblies (modules, printed circuit boards, and chassis-mounted parts). Corrective maintenance time includes interchange, reassembly, alignment, and checkout for all corrective actions.

3.6 Workmanship. Workmanship shall be as specified in 3.6.1 and 3.6.2.

3.6.1 General workmanship. Workmanship shall be as specified in the Workmanship paragraph of MIL-E-16400.

3.6.2 Workmanship screen. ACUs shall withstand a defect detection vibration screen (see 4.9.1) of random type vibration at $0.04g^2$/Hz $\pm$3dB from 80 Hz to 350 Hz and temperature cycling as specified in 4.9.2.

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TABLE IV. Test provisions.

Function under test	Signals inserted	Item under test	Minimum coupling level
VLF/LF	10 kHz to 160 kHz	VLF/LF antenna for probe test from J-box insertion point to J-box output	-60 dB
MF/HF	2.5 MHz to 20 MHz	MF/HF antenna for probe test from insertion point1/ in VLF lines in antenna interface system to output of ACU	-70 dB
MF/HF	2.5 MHz to 20 MHz	MF/HF transmission line from J-box insertion point to output of ACU	-40 dB
VHF/UHF	150 MHz to 400 MHz	VHF/UHF antenna for probe test from insertion point1/ in VLF lines in antenna interface system to output of ACU	-80 dB
VHF/UHF	150 MHz to 400 MHz	VHF/UHF transmission line from J-box insertion point to output of ACU	-40 dB

1/ Insertion of MF/HF and VHF/UHF signals in the VLF lines shall be by means of a balanced input to the forward and after transmission line pair of the quad coaxial line. This insertion to be made in the antenna interface system (AIS).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied; b) Government product inspection to measure quality of the product to be offered for acceptance; and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

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4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be defined in MIL-STD-109.

4.2 Classification of inspections. The inspection requirements specified herein are classified as specified in a through c.

a. First article inspection (see 4.3)
b. Quality conformance inspection:
   1. Production inspection (Group A) (see 4.4.1)
   2. Production control inspection (Group B) (see 4.4.2)
   3. Environmental inspection (Group C) (see 4.4.3)
c. Inspection of preparation for delivery (see 4.10)

4.3 First article inspection. Unless otherwise specified (see 6.2), one antenna group shall be required for first article inspection. First article inspection shall consist of all examinations and testing to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE V.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in 4.4.1 through 4.4.3.2.

4.4.1 Production inspection (Group A). Production inspection shall be made on every antenna group offered for delivery. Production inspection shall comprise such examinations and tests which will prove the workmanship and reveal omissions and errors of the production process, such as functional and performance tests at a limited number of points in the required range, tests which detect deviations from design, tests of controls and adjustments, and tests which detect hidden defects of materials. Production inspection shall include the examinations and tests specified in Group A of TABLE V.

4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis as specified in 4.4.2.1. Production control inspection shall be in addition to the production inspection specified in 4.4.1, and shall consist of examinations and tests which encompass functional and performance tests throughout the entire range of operation, and tests to determine deviations in the processes and equipment required to fabricate the antenna group. These tests shall be performed on the complete antenna group as offered for delivery. Production control inspection shall include the examinations and tests specified in Group B of TABLE V.

4.4.2.1 Sampling plan for Group B inspection. Antenna groups submitted for Group B sampling tests shall be selected by the contractor under the supervision of the Government Quality Assurance Representative (QAR) and shall be representative of current production. Group B inspection, including sampling, shall conform to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. Group B inspection shall be performed on inspection lots that have passed Group A inspection specified in 4.4.1. The acceptable quality level shall be 6.5 percent defective for each of the Group B examinations and tests, and the inspection level shall be S-3 for normal, tightened, and reduced inspection.

4.4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be accomplished on a sampling basis as specified in 4.4.3.1. Environmental inspection shall encompass environmental tests to prove the durability of the materials, parts, units, and the antenna group as a whole; tests of the effects of changes of environment (such as, extremes of temperature and humidity); and tests of the effects of shock and vibration. Environmental inspection shall include the examinations and tests specified in Group C of TABLE V.

4.4.3.1 Sampling for environmental inspection. One antenna group shall be selected from each successive 10 antenna groups, or fraction thereof, produced. The first sample antenna group shall be selected from the first month's production.

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TABLE V. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Workmanship screen	3.6.2	4.9	X	X		
Reliability	3.4	4.7	X			
Maintainability	3.5	4.8	X			
Shock	3.2.1.2	4.5.1.1				
Radome and antenna unit			X			
Control unit			X			X
Vibration	3.2.1.3	4.5.1.2				
Radome and antenna unit			X			
Control unit			X			X
Noise	3.2.1.7	4.5	X			X
EMP	3.2.1.8	4.5.1.3	X			
Mercury and radioactive materials	3.2.3.2.1	4.5.1.4	X	X		
SUBSAFE certification	3.2.3.2.2	4.6	X	X		
Primary power	3.2.4.1	4.5	X		X	
Insertion loss	3.2.4.2	4.5.1.5	X	X		
EMC	3.2.4.3	4.5.1.6	X			
Enclosure	3.2.5.2	4.5	X		X	
Magnetic permeability	3.2.5.3	4.5	X			
Size	3.3.1.2, 3.3.8.3	4.5	X	X		
Weight	3.3.1.4, 3.3.8.4	4.5	X	X		
Hydrostatic pressure	3.3.1.3, 3.3.9.2, 3.3.11.1	4.5.1.7	X	X		
VSWR	3.3.2.1c, 3.3.4.1c, 3.3.5.1c, 3.3.6.1c	4.5.1.8	X	X		
Pattern	3.3.2.1d, 3.3.4.1d, 3.3.5.1d, 3.3.6.1d	4.5.1.9	X		X	
Power input	3.3.2.1e, 3.3.4.1e, 3.3.5.1e, 3.3.6.1e	4.5.1.10	X	X		
Bandwidth	3.3.2.1f	4.5.1.11	X		X	
Efficiency	3.3.2.2	4.5.1.12	X		X	
Effective height	3.3.3.1b	4.5.1.13	X		X	
Decoupling isolation	3.3.3.1.1	4.5.1	X		X	
Loop Q	3.3.3.1.2	4.5.1	X		X	
Loop inductance	3.3.3.1.3	4.5.1	X		X	
Depth transducer output	3.3.3.2	4.5.1	X		X	
Depth transducer pressure	3.3.3.2.3	4.5.1	X			
VLF preamplifier gain	3.3.3.3.1	4.5.1	X		X	
VLF frequency response	3.3.3.3.2	4.5.1.14	X	X		
VLF phase difference	3.3.3.3.3	4.5.1	X		X	
VLF phase shift variation	3.3.3.3.4	4.5.1.15	X		X	
VLF harmonic distortion	3.3.3.3.5	4.5.1	X	X		
VLF phase characteristics	3.3.3.3.6	4.5.1	X	X		
VLF/LF sensitivity	3.3.3.3.7	4.5.1	X	X		
Antenna gain	3.3.4.1f, 3.3.5.1f, 3.3.6.1f	4.5.1.16	X	X		
Polarization	3.3.4.1g, 3.3.5.1g, 3.3.6.1g	4.5.1.17	X		X	
RF feed system losses	3.3.4.2, 3.3.5.2, 3.3.6.2	4.5.1	X	X		
NAVSAT phase shift variation	3.3.4.4a	4.5.1	X		X	
VHF/UHF preamplifier noise figure	3.3.7	4.5.1	X	X		

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TABLE V. Examinations and tests (continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
VHF/UHF preamplifier frequency response	3.3.7	4.5.1	X	X		
VHF/UHF preamplifier gain	3.3.7	4.5.1	X	X		
VHF/UHF preamplifier compression	3.3.7	4.5.1	X	X		
Cable cold bend test	3.3.9	4.5.1	X	(for each cable lot)		
Cable attenuation	3.3.9.1	4.5.1	X	X		
Hydraulic shock	3.3.9.3, 3.3.11.3	4.5.1	X		X	
Underwater explosion	3.3.11.2	4.5.1	X			
Control provisions	3.3.12.1	4.5.1	X		X	
Monitor provisions	3.3.12.2	4.5.1	X		X	
Signal insertion	3.3.12.3	4.5.1	X		X	
Coupling level	3.3.12.3.1	4.5.1	X		X	
Electrical interface	3.3.12.5	4.5.1	X		X	
General workmanship	3.6.1	4.5	X	X		
Simultaneous operation	3.3.1	4.5.1	X		X	
Surface examination	3.2.3	4.5	X	X		
Painting	3.2.3.3	4.5	X	X		
Marking	3.1	4.5	X	X		
Humidity	3.1	4.5	X			X
Temperature	3.2.1.1	4.5	X			X
Inclination	3.2.1.4	4.5	X			X
Safety	3.2.6	4.5	X		X	

4.4.3.2 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the QAR the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials, processes, and so forth. If the QAR does not consider that the corrective action will enable the product to conform to specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.4.4 Reinspection of conforming production control and environmental sample units. Unless otherwise specified (see 6.2), sample units which have been subjected to and have passed production control (Group B), or environmental (Group C) inspections, or both, may be accepted on the contract, provided all damage has been repaired and the antenna groups are resubjected to, and pass, the production inspection (Group A).

4.5 Test methods. Unless otherwise specified herein, the examinations and tests listed in TABLE V should be performed in accordance with the test methods specified in MIL-E-16400.

4.5.1 Performance test. The antenna group shall be subjected to a performance test to determine compliance with the performance parameters and safety requirements herein, except as specified in 4.5.1.1 through 4.5.1.17.

4.5.1.1 Shock. The antenna group equipment shall be hard mounted to the shock table by a test fixture and subjected to the shock inputs of MIL-S-901 (as modified by the APPENDIX) for Grade A, Type A, Class I equipment. The antenna group shall be mounted to the test fixture in the same way that the antenna group is to be mounted in the ship. The medium weight, high impact shock table shall be used for this test. The equipment shall be energized and in operating condition after each shock blow. Insertion loss and VSWR of the antenna shall be measured before and after the test on the antenna and radome unit. Operation of the antenna controls and indicators shall be verified before and after the test on the ACU.

4.5.1.2 Vibration. The antenna group equipment shall be hard mounted to the vibration table by a test fixture and subjected to the amplitude inputs of MIL-STD-167-1, the vibration displacement of environmental vibration table for each of three orthogonal axes (principal axis of units). The frequency range shall be 4 Hz to 33 Hz. The ACU shall be bolted to the test fixture in the same way that the ACU will be mounted to the rack. The resonant frequency of the fixture shall be greater than

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75 Hz. Insertion loss and VSWR of the antenna shall be measured before and after the test on the randome and antenna unit. Operation of the ACU controls and indicators shall be verified before and after the test on the ACU.

4.5.1.3 EMP. Tests specified in 4.5.3.1 through 4.5.3.2.4 shall be conducted to determine compliance with the requirements of 3.2.1.8.

4.5.1.3.1 External equipment. The requirements of 3.2.1.8.1 shall be verified by an analysis or test of a pulse induced from a simulated exo-atmospheric nuclear explosion whose RF field of the form and amplitude specified in the following equation:

$$E(t) = E_o(e^{-\alpha t} - e^{-\beta t}) \text{ and } E_o(e^{-\alpha t} - e^{-\beta t})$$

$$H(t) = \frac{E(t)}{Z_o}, \text{ where}$$

E(t) = Electric field in V/m at time, t

H(t) = Magnetic field in A per meter at time, t

$E_o = 5.25 \times 10^4$ V/m

$= 4.0 \times 10^6$ per second

$= 4.76 \times 10^8$ per second

$Z_o = 377$ ohms.

t = Time in seconds

4.5.1.3.2 Inboard equipment. The requirements of 3.2.1.8.2 shall be verified by the test method specified in 4.5.1.3.2.1 through 4.5.1.3.2.4.

4.5.1.3.2.1 Applicable test frequencies. The tests shall be performed for the injection frequencies of 0.01 MHz, 0.1 MHz, 0.5 MHz, 1 MHz , 2 MHz, 5 MHz, 10 MHz, 20 MHz, 50 MHz, and 100 MHz. The test shall be performed on the critical frequencies of the antenna group, such as local oscillator, power switching frequencies and harmonics, RF frequencies, clock frequencies, bandpass frequencies, and so forth as specified in the electromagnetic interference test plan.

4.5.1.3.2.2 Apparatus. Test apparatus shall consist of the equipment in a through d:

a. Source injection pulse generator with the capabilities specified in 1 through 4:

1. Damped sinusoidal transient as shown in the applicable limit and as described by the equations in (a) and (b):

(a) $I_{pin}(t) = 1.05\ I(f)\ e^{-0.2ft}\ SIN\ (2\ ft)$

(b) $V_{pin}(t) = 1.05\ V(f)\ e^{-0.2ft}\ SIN\ (2\ ft)$

where,

$I_{pin}(t)$ = Pin to case current, in A at time, t

$V_{pin}(t)$ = Pin to case voltage, in V at time, t

f = Specific test frequency, in MHz

t = Time in seconds

I(f) = Current determined from applicable limit curve of FIGURE 2

V(f) = Voltage determined from applicable limit curve of FIGURE 2

2. Variable amplitude so the injected transient can be increased from a minimum to a maximum specified level (that is, at least 10 percent, 50 percent, and 100 percent of the amplitude shall be tested at each specified test frequency)

3. Frequency range capable of generating the required test frequencies

4. Current output level of at least 10 A into a 100 ohms load

b. Current probe

c. Oscilloscope with a minimum bandwidth capability of 100 MHz

d. Interconnecting test cable

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4.5.1.3.2.3 Test setup and procedure. The test setup and procedure shall be as specified in 4.5.3.2.4. A functional analysis shall be performed before the certification testing to identify the items in a through d:

a. The interface pin connectors
b. The test sample functions which shall be monitored to determine system failure or transient upset
c. The additional test frequencies (that is, interface cable resonant frequency, clock frequencies, data rates, and so forth)
d. Test setup shall be as shown in FIGURE 3.

4.5.1.3.2.4 Test procedure. The test procedure shall be as specified in a through e:

a. Apply the test signal to each interface pin of the test sample, at each test frequency.
b. Monitor the test sample functions in all modes of operation to determine whether there is a failure or transient upset.
c. Inject one transient pulse on each interface pin and at each test frequency.
d. The test sample shall be tested with power off and on.
e. Determine test frequency and level at which a failure or upset occurs.

NOTES:

The test sample shall be tested on a ground plane. There shall be no insulated materials between the test setup and the ground plane.

The test cable running between the test sample and the damped sinusoidal generator shall be kept 4 cm to 6 cm above the ground plane.

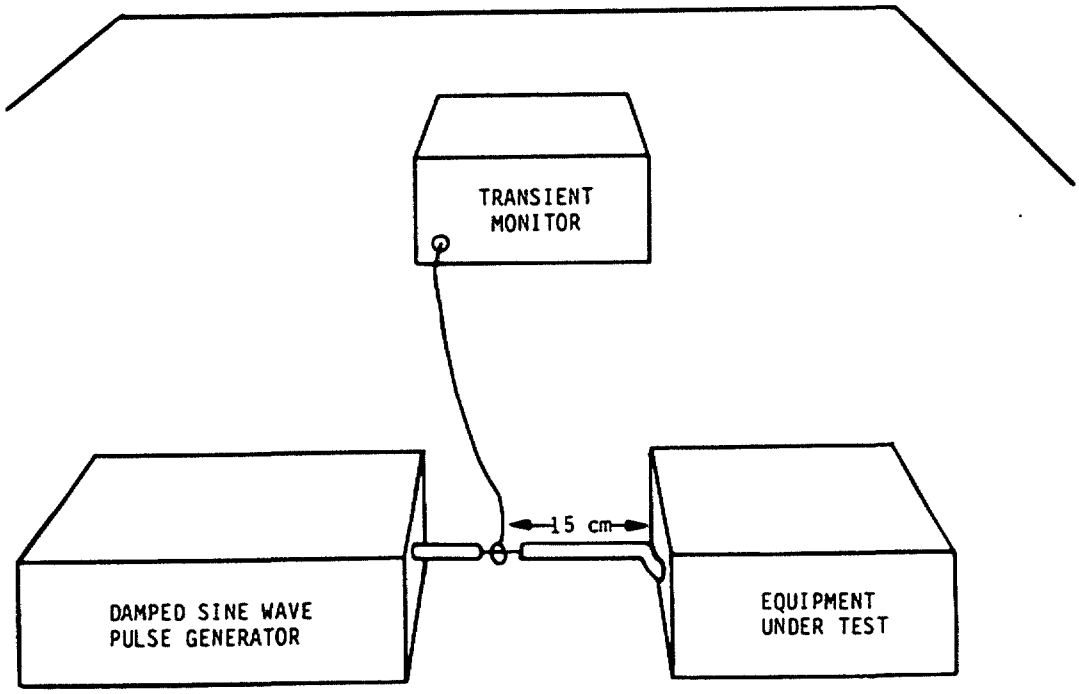


FIGURE 3. EMP test setup.

4.5.1.4 Mercury contamination. Antenna groups that have passed all required examinations and tests and are ready for packaging shall be tested for mercury contamination. The tests specified in a through c shall be conducted to determine whether contamination of mercury exists:

a. The equipment and packaging material shall be enclosed in a polyethylene bag or close-fitting, air-tight container and placed in an oven at 52 °C ±2.8°C (125 °F ±5 °F) for 1 hour.
b. The trapped air in the bag shall be sampled and measured for metallic mercury contamination. If the mercury vapor contamination exceeds 0.01 milligram per cubic meter, the equipment shall be rejected.

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c. Mercury vapor concentration may be determined with a mercury vapor detector, such as a portable General Electric Vapor Detector (catalog number 8257557G-3), Beckman Instruments Model K-23, or equal instruments having equivalent ranges and capabilities. Certain vapors, such as benzene, interfere with this type of mercury vapor detector.

4.5.1.5 Insertion loss. Tests to demonstrate compliance with requirements of 3.2.4.2 shall be conducted at 150 MHz, 300 MHz, and 400 MHz.

4.5.1.6 EMC. The requirements specified in 3.2.4.3 shall be verified by tests performed in accordance with MIL-STD-462. The tests shall employ sweep frequency spectrum analysis and provide annotated data plots.

4.5.1.7 Hydrostatic pressure. Each radome-antenna unit, outboard cable assembly, and hull penetration insert shall be submitted to a hydrostatic pressure test to demonstrate compliance with 3.3.1.3, 3.3.9.2, and 3.3.11.1, as applicable.

4.5.1.7.1 Vacuum leak. Following the hydrostatic test of 4.5.7 on the radome-antenna unit and stabilization of the unit to room temperature, the unit shall withstand a partial vacuum of 71 cm of mercury for a minimum period of 2 hours. Indicated pressure readings taken 15 minutes after evacuation and at the end of the test period shall be compared. The deviation shall be within 10 percent.

4.5.1.8 VSWR. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.2.1c, 3.3.4.1c, 3.3.5.1c, and 3.3.6.1c. Measurements shall be made at 1 kHz increments from 2 MHz to 3.0 MHz, 2 kHz increments from 3.0 MHz to 7.0 MHz, 0.5 kHz increments from 7.0 MHz to 14 MHz, and 1 MHz increments from 14.0 MHz to 30.0 MHz. Swept frequency measurements shall be made from 225 MHz to 400 MHz and 950 MHz to 1150 MHz.

4.5.1.9 Pattern. Tests shall be conducted to demonstrate compliance with the requirements 3.3.2.1d, 3.3.4.1d, 3.3.5.1d, and 3.3.6.1d in both horizontal and vertical planes as specified.

4.5.1.10 Power input. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.2.1e, 3.3.4.1e, 3.3.5.1e, and 3.3.6.1e. MF/HF antenna power input should be maintained at 1000 W for 5 minutes at each of four frequencies (one between 2 MHz and 4 MHz, one between 4 MHz and 12 MHz, one between 14 MHz and 20 MHz, and one between 20 MHz and 30 MHz). VHF/UHF power input should be maintained at 100 W for 15 minutes at each of four frequencies between 225 MHz and 400 MHz. IFF power input should be maintained at 100 W for 15 minutes at each four frequencies between 950 MHz and 1150 MHz.

4.5.1.11 Bandwidth. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.2.1f. Bandwidth shall be measured at 2.25 MHz, 4 MHz, 14 MHz, and 30 MHz in both HF only and multifunction modes.

4.5.1.12 Efficiency. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.2.2. Efficiency measurements shall be conducted at 2 MHz, 2.4 MHz, 3 MHz, 5 MHz, 20 MHz, and 30 MHz.

4.5.1.13 Effective height. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.3.1b. The effective height of the VLF antenna shall be measured at 20 kHz and 120 kHz.

4.5.1.14 VLF frequency response. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.3.3.2. Measurements shall be made at 10 kHz and 14 kHz plus at increments of 20 kHz from 20 kHz to 160 kHz.

4.5.1.15 VLF phase shift variation. Tests shall be conducted by applying a 100 kHz input signal through the VLF antenna while the radome and antenna unit is within an environmentally controlled space. The VLF output from the radome-antenna unit shall be terminated in 50 $\pm$j0 ohms.

4.5.1.16 Gain. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.4.1f, 3.3.5.1f, and 3.3.6.1f. Gain measurements shall be made at 150 MHz, 225 MHz, 300 MHz, and 400 MHz for the VHF/UHF antenna, 950 MHz and 1150 MHz for the IFF antenna, and 245 MHz and 320 MHz for the SATCOM antenna.

4.5.1.17 Polarization. Tests shall be conducted to demonstrate compliance with the requirements of 3.3.4.1g, 3.3.5.1g, and 3.3.6.1g. Tests shall be made at one point within the specified frequency range.

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4.6 SUBSAFE requirements and certification. SUBSAFE requirements and certification shall be as specified in 4.6.1 through 4.6.2.5.

4.6.1 Glossary of terms. The definitions in 6.4.2 through 6.4.6 are included to clarify the certification requirements.

4.6.2 Welding. Welding shall be as specified in 4.6.2.1 through 4.6.2.5.

4.6.2.1 Electrodes. Quantitative test reports of the chemical and mechanical properties of electrodes conforming to MIL-E-22200, shall be submitted for approval.

4.6.2.2 Procedures. Written procedures covering all welding processes utilized in manufacture of the hull fitting in accordance with MIL-STD-278 shall be submitted for approval to the Naval Ship Systems Engineering Station, Code 06, Philadelphia, PA 19112. Welding procedures which have not been previously approved by NAVSSES shall be submitted for approval with the results of the welding procedure qualification as specified in the Welding procedure qualification paragraph of MIL-STD-248.

4.6.2.3 Qualification. Procedures in effect (approved by NAVSSES) and qualification records of each individual welder performing welding shall be in accordance with MIL-STD-248.

4.6.2.4 Joint design. Design of welded joints shall be in accordance with MIL-STD-278 and MIL-STD-22. Joint design designation shall be designated on each drawing. Joint designs shall be designated in the weld procedure.

4.6.2.5 Nondestructive testing (NDT). All welds in primary and secondary boundaries require dye penetrant or magnetic particle inspection. In addition to dye penetrant or magnetic particle inspection, all full penetration welds shall have 100 percent radiographic inspection. All inspection shall be in accordance with MIL-STD-271.

4.7 Reliability qualification test. The first article antenna group (including all units) shall be subjected to a fixed length qualification test in accordance with MIL-STD-781. The fixed length Test Plan XXIC shall be utilized. The operational duty cycle and test conditions shall be specified 4.7.1 through 4.7.7.1.

4.7.1 Duty cycle. The antenna group undergoing reliability testing shall operate continuously, except for a shutdown period of 4 hours during every 168 hours of testing. The shutdown shall occur when the test temperature is approaching and at 0° Celsius (C).

4.7.2 Power testing. During reliability testing, the antenna group shall be subjected to 1 minute of specified MF/HF, UHF, and IFF power transmissions every 24 hours.

4.7.3 Temperature cycling. The antenna group shall be subjected to repetitive temperature cycling. A temperature cycle shall consists of 4 hours approaching and at 0°C followed by 4 hours approaching and at 50°C.

4.7.4 Vibration cycling. The antenna group shall be subjected to continuous repetitive sine sweep vibration with frequency range of 4 Hz to 33 Hz to 4 Hz. The vibration amplitude shall be 0.58 mm $\pm$ 0.102 mm (0.020 in. $\pm$ 0.004 in.). The time to sweep the frequency up and down shall be 10 minutes $\pm$2 minutes. There shall be no vibration cycling during the duty cycle shutdown period.

4.7.5 Voltage cycling. Input voltage shall be maintained at 110 percent of nominal for the first 8-hour test temperature cycle, at the nominal value for the second test cycle, and at 90 percent of nominal for the third test cycle. The cycling procedure is to be repeated continuously throughout the test.

4.7.6 Performance measurement. During reliability testing VSWR and insertion loss measurement within all frequency bands shall be made every 24 hours.

4.7.7 Failure. The procuring activity shall be notified of failures during reliability testing (6.4). The procuring activity shall classify failures in accordance with the Failure actions paragraph of MIL-STD-781. When two or more replaceable assemblies fail simultaneously, each shall be counted as a separate failure, even though one or more of the failures may be dependent.

4.7.7.1 Post-testing equipment disposition. At the completion of reliability testing, the tested antenna group shall be refurbished as directed by the procuring activity for delivery as production or training units.

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4.8 Maintainability demonstration. The first article antenna group shall be subjected to a maintenance capability and diagnostics demonstration at the organizational level. The demonstration shall be performed by qualified technicians and shall verify that the M_{ct} requirement specified in 3.5 is satisfied.

4.8.1 Demonstration conditions. The maintainability requirements shall be demonstrated in accordance with MIL-STD-471, Test Method 9.

4.8.2 Demonstration environment. The demonstration shall be performed in an environment similar to that in which the antenna group will be installed.

4.8.3 Acceptance or rejection. The maintainability demonstration shall exhibit that the antenna group conforms to the accept criteria prior to delivery. Failure to conform to the requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified. The contractor shall, at no additional cost to the Government:

a. Develop an approach for redesign or correction of all deficiencies.
b. Upon approval of this approach, the contractor shall repeat the demonstration until an accept decision is reached.

4.9 Workmanship screen. Vibration and temperature cycling tests performed on each ACU. Vibration shall be performed prior to temperature cycling. The vibration may be performed at the module, drawer, or end item level. All the hardware, including cables and connectors, shall be exposed to vibration.

4.9.1 Random vibration. The vibration shall be random or, subject to procuring activity approval, pseudo-random or complex waveform vibration, for an accumulated time of 10 minutes in the axis deemed most susceptible to vibration excitation. ACUs shall be hard-mounted (without shock isolators) and subjected to the vibration conditions of FIGURE 4. Input vibration levels shall be measured at the mounting points of the item under vibration. If variations are found at these points, the level used for control purposes shall be the average of the levels at the mounting points. Control equipment having a bandwidth no greater than 10 Hz for vibration frequencies up to 500 Hz, and 100 Hz for vibration frequencies above 500 Hz, shall be used for the control and analysis of the acceleration spectral density (ASD). The instantaneous acceleration peaks shall be limited to three times the acceleration level. The ACU shall be energized during vibration and appropriate input signals applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during screening shall be corrected and the vibration resumed.

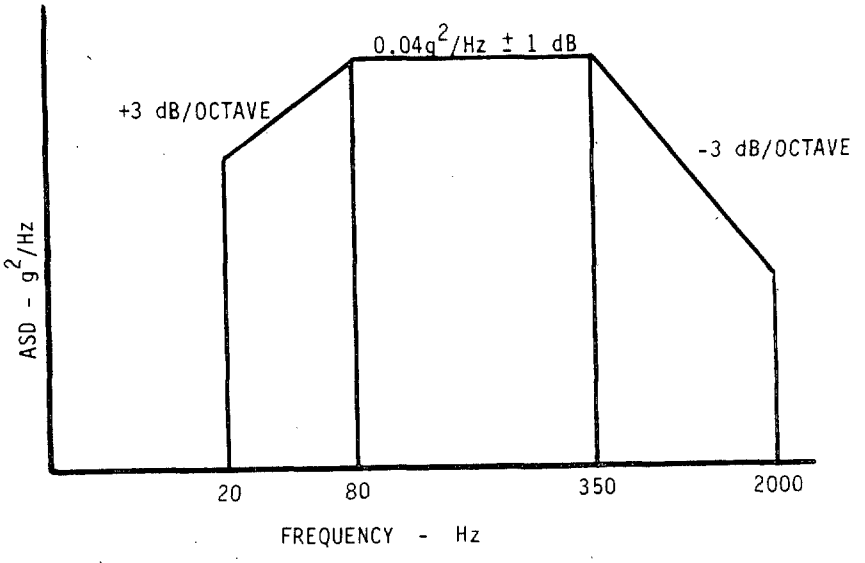


FIGURE 4. Random vibration curve.

4.9.2 Temperature cycling. Each ACU shall be subjected to 10 cycles of the temperature curve shown in FIGURE 5. The temperature rate of change shall be not less than 5° C per minute. Power shall be turned on and off at the indicated times. The ACU shall be positioned for maximum exposure to the changing temperature. Where performance measurements are specified, a minimal functional

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operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the coupler reaches 80 percent of the chamber temperature. When failures occur, the ACU shall be reworked and the cycling continued for a cumulative total of 10 cycles.

4.10 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

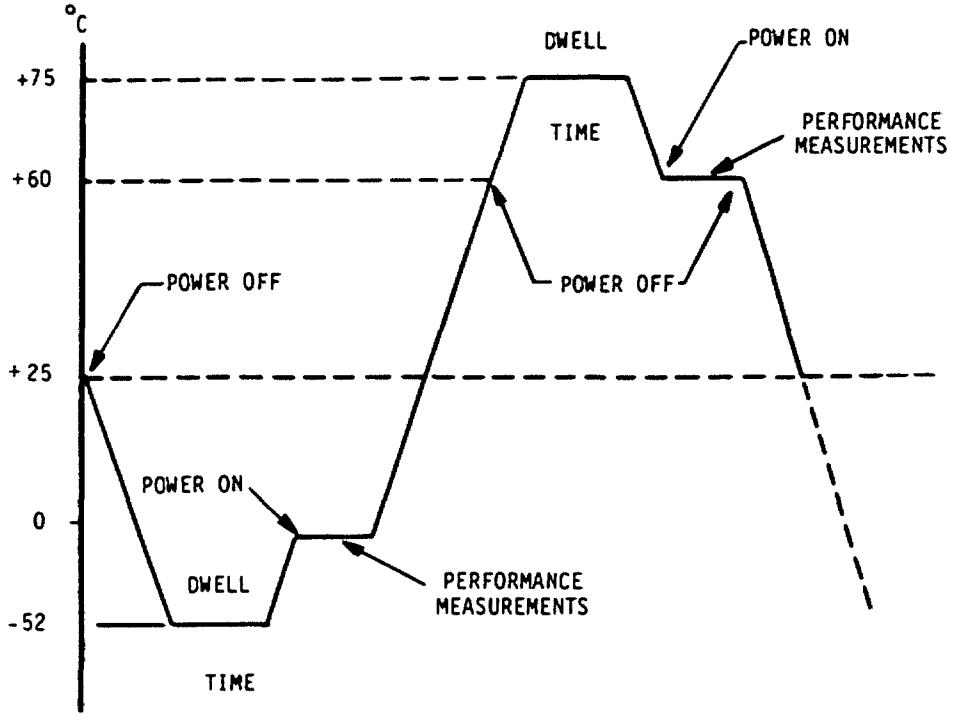


FIGURE 5. One cycle of temperature curve (not to scale).

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna group covered by this specification is intended for use on TRIDENT class submarines.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Number of first article samples to be submitted if other than specified in 4.3
c. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 52.209-3 of the Federal Acquisition Regulations. The first article shall consist of one antenna group. The first article should be a first production item. The contracting

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officer should include specific instructions in all procurement instruments regarding arrangements for examinations, tests, and approval of the first article.

6.4 Definitions. Definitions of terms used in this specification are given in 6.4.1 through 6.4.6.

6.4.1 Failure. Failure is defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements specified herein.

6.4.2 Pressure hull structure. Pressure hull structure is defined as all structure subject to the differential of sea pressure to internal hull pressure, including pressure hull envelope and primary boundary.

6.4.3 Penetrations. Penetrations are defined as structural compensation parts such as sleeves, hull valve bodies, antennas, and trunks which pass through and are welded to the pressure envelope, closure bulk-heads, hard tanks, and other structures subject to the differential of sea pressure to internal hull pressure.

6.4.4 Fittings. Fittings are defined as all items and assemblies which are attached either mechanically or are welded to pressure hull penetrations, pressure hull inserts, or pressure hull inserts, or pressure hull plating. This includes structural piping, mechanical, tubes, antennas and electrical pressure hull fittings, and mechanical stuffing boxes subject to primary and secondary boundaries.

6.4.5 Primary boundary. The primary boundary is defined as any material which excludes sea pressure at all times.

6.4.6 Secondary boundary. The secondary boundary is defined as any material which excludes sea pressure in event primary boundary material fails.

Preparing activity:
NAVY-EC
(Project No. 5985-N516)

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APPENDIX

SUBMARINE SHOCK REQUIREMENTS

10 SCOPE

10.1 This APPENDIX is applicable when invoked by procurement documents. This APPENDIX modifies the shock test requirements contained in MIL-S-901C to make them compatible with approved specifications applicable to submarines. This APPENDIX also provides the requirements (compatible with approved submarine specifications) for underwater explosion (UNDEX) testing of submarine hull penetrations.

20 APPLICABLE DOCUMENTS

20.1 The following document forms a part of this APPENDIX to the extent specified herein.

SPECIFICATION

MILITARY

MIL-S-901C	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For

30 REQUIREMENTS

30.1 MIL-S-901C shall be modified as follows:

Page 1, paragraph 1.1 Scope., last sentence: Delete period, substitute comma, and add "and more particularly to shock induced in ships by noncontact underwater explosions."

Page 2, under "MILITARY STANDARDS": Add "MIL-STD-798 Nondestructive Testing, Welding, Quality Control, Material Control and Identification and HI-Shock Test Requirements for Piping System Components for Naval Shipboard Use".

Page 2, under "PUBLICATIONS": Delete "NAVSHIPS 250-423-30 and NAVSHIPS 250-423-31" and add: "NAVSHIPS INSTRUCTION 9690.7" "Location of Class HI Shock Test Facilities", and: "Design Data Sheet DDS 072-1" "Shock Design Values (Confidential)". Add new subtitle "SUPSHIP THREE" and below that add: "Report SUPSHIP 280-8 of May 1973" "Shock Design Criteria for TRIDENT Submarine".

Page 2, under "NAVAL RESEARCH LABORATORY": Add "Report 7366" "Shipboard Shock and Navy Devices for its Simulation" "NRL ltr report 6260-7A:ROB:kep of 18 Feb 1964".

Page 3, add new paragraph 3.1.2.1.1:

"3.1.2.1.1 Hull mounted (external). External hull mounted items are sonar transducers, antenna housings, and other appendages mounted outside the pressure hull."

Page 4, paragraph 3.1.4.1, second line: Delete "applicable...4.2.4.1.2)." and substitute "for equipment which together with the test fixture weighs no more than 550 pounds."

Page 5, paragraph 3.1.4.2, second line: Delete "approximately 250 pounds to 6000 pounds." and substitute "up to approximately 6000 pounds."

Page 5, paragraph 3.1.4.3: Delete in its entirety and substitute:

"3.1.4.3 Heavy weight. The heavy weight test is applicable to items weighing up to approximately 60,000 pounds in operating condition (see 4.2.3.3, 4.2.4.3, and FIGURE 13); and to external hull-mounted equipment (see 3.1.2.1.1 and 4.2.4.4)."

Page 5, paragraph 3.1.5.3, after last sentence add: "If subassemblies associated with a principal unit or subsidiary component are not available at the time the principal unit or subsidiary component is scheduled to be tested, they may be represented in a manner acceptable to the Naval Electronic Systems Command (NAVELEX), for the test. If subassemblies are not tested as part of a principal unit or subsidiary component, they must be tested separately (see 4.2.2.3)."

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APPENDIX

Page 6, paragraph 3.2.2: Delete in its entirety and substitute:

"3.2.2 Extension of shock tests approval. The shock test approval achieved by test of an item may be extended to cover an identical or similar item, provided that it can be clearly shown that the original item was shock-tested and approved in accordance with this specification, and that the design of the untested equipment, its intended services, and shipboard installation are such as to result in equal to or better degree of shock resistance. Shock test extensions are subject to approval by the designated approval authority. The following requirements apply to extension of shock test approval:

(a) Shock test extension policies apply to items identical to previously tested and approved designs, to items identical to those previously approved on the basis of shock test extension, and to items similar to previously approved designs.

(b) In cases where the design of an item is identical to that of a previously shock-tested and approved item, records of the shock test extension shall reference the original shock test report (plus any post-test reports), and shall certify that any intended variation of service or installation relative to that represented during the original shock test will not result in lessened shock resistance and shall reference Government correspondence which originally approved the item from a shock standpoint. If the above is not available for previously tested items, reference the shock test report (and any post-test reports) and identify the facility which conducted the test.

(c) In cases where the design of an item as identical to that of an item which was previously approved on the basis of shock test extension, records of the additional extension to new applications shall provide the information indicated in 3.2.2(b) and shall also reference the original extension approval.

(d) In cases where the design of an item is not identical to that of a previously approved item, or in cases where intended variation of service or installation are significant from a shock standpoint, the request for extension of approval must be accompanied by the following supporting evidence, as appropriate:

(1) Detailed drawings of tested and untested items and delineation of orientation of the items as installed in the ship.

(2) A copy of the report of the shock test and post shock test reports upon which the requested extension is based.

(3) A detailed comparison of the differences in materials and design showing that the untested item on its foundation has equal or greater shock resistance than the tested item on its test foundation.

(e) Extension will not be granted for equipments manufactured by different manufacturers unless specific Government approval is obtained.

(f) In case of doubt, general guidance as to allowable variations in component characteristics between tested and untested items may be obtained from the Government approval authority.

(g) Shock tests of valves may be extended to other sizes of the same design in accordance with criteria as specified in MIL-STD-798.

(h) Conditional shock test extensions may be granted for heavyweight items prior to the test of the item upon which the extension is to be based. In such cases, (b), (d)(1), (d)(3), (e), (f), and (g) above shall apply. A final shock test extension will be granted when a shock test report is submitted which certifies that the item tested has satisfactorily passed the shock test. The vendor shall, at his expense, make similar necessary corrections to the extended item as may be required to be made to the tested item due to the results of the shock test.

(i) Subject to conforming to the requirements of (d)(1),(d)(2),and (d)(3) above, items such as those listed below whose degree of similarity is difficult to assess may be extended from tests of generally similar components by reference to calculations furnished to show that the tested item and the untested item are designed to the same criteria and that the untested item is at least equal in shock resistance to the tested item. Typical items in this category include piping system castings, ventilation system fittings, nonbuilt-in tanks, ladders, gratings, floor plates, handrails, tool stowage attachments, and repair parts stowage attachments."

Page 6, paragraph 3.2.2.1, first line: Delete "Items" and substitute "Medium weight items".

Page 6, paragraph 3.2.2.1, after last sentence: Add "Subject to meeting the requirements of paragraph 3.2.2(d)(1), (d)(2), (d)(3):

(a) All items subjected to the lightweight test and previously qualified to any issue of MIL-S-901 shall be considered to be qualified under MIL-S-901C without any extension action.

(b) All items subjected to the medium weight test and previously qualified to any issue of MIL-S-901 shall be considered to be qualified under MIL-S-901C upon successful completion of the additional test (at 30 degrees or 90 degrees inclination) required under paragraph 4.2.4.2. (All such items previously qualified with shock blows imposed along two different principal axes of the item shall be considered to be qualified under MIL-S-901C without additional testing)."

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APPENDIX

Page 6, paragraph 3.3.1, line 3: After "assembly" add "or arrangement".

Page 6, paragraph 3.3.1(b): After "letter" add "(if applicable), shock qualification extended from test report number ____" "Approved by_____".

Page 7, paragraph 4.2.1: Add "NAVSHIPS INSTRUCTION 9690.7 lists the approved Class HI shock test facilities for use in connection with this specification. Machines not listed therein must be inspected by the cognizant Government authority and certified to NAVSEA to be constructed and installed in accordance with the drawings referenced herein prior to acceptance of equipment tested on those machines."

Page 7, paragraph 4.2.1: Delete "equipment" in four places and substitute "tests."

Page 7: Add new paragraph 4.2.1.1:

"4.2.1.1 Platforms. Lightweight equipment may be tested on the medium weight machine or the heavyweight platform. Medium weight equipment may be tested on the heavyweight platform."

Page 7, paragraph 4.2.2.1: In first line, after "shall" add "be mounted"; in line 3, delete "can" and substitute "will"; in line 4, insert "as" between "be" and "specified"; and in line 5 after "individual equipment specification" add "or the contract".

Page 7, paragraph 4.2.2.2, line 5: Delete "provide a fixture...principal unit." and substitute "mount the equipment to simulate an equal or more severe condition of installation in service."

Page 7, paragraph 4.2.3.1: Delete "equipment" in two places and substitute "tests".

Page 8, paragraph 4.2.3.3: Delete "equipment" and substitute "test (internal equipment)" in heading; in line 6, delete "30,000 pounds (40,000 pounds at the San Francisco Naval Shipyard Facility)" and substitute "60,000 pounds (including the foundation and/or test fixture.)".

Page 8, after paragraph 4.2.3.3: Add new paragraph 4.2.3.4:

"4.2.3.4 Heavyweight test (external equipment). External hull mounted equipment shall be mounted to the underside of the floating shock platform near the edge facing the charge. The mounted bracket shall approximate the structure used for shipboard mounting the equipment."

Page 9, paragraph 4.2.4: Add: "This specification is very general so as to cover the entire field of shipboard equipment and systems. In order to apply this specification properly and to avoid post-test disagreements, test procedures shall specify, describe, or define all the features enumerated in 6.1. For heavyweight tests only the test procedure shall be approved by the procuring activity and the Government review and acceptance authority prior to the shock test. If retest of heavyweight items is required, the extent of retest to verify design modifications shall be as approved by the review and acceptance authority (see 4.2.7.2)."

Page 9, paragraph 4.2.4.1: Delete "equipment" and substitute "test".

Page 9, paragraph 4.2.4.1.2: Delete in its entirety.

Page 9, paragraph 4.2.4.2: Delete "equipment" and substitute "test".

Page 9, paragraph 4.2.4.2: Delete first sentence.

Page 9, paragraph 4.2.4.2: In second sentence after "as specified in" add "the"; in the sixth sentence after "fixture used" add "for the included orientation", and after "10-2" add "or 10-3".

Page 9, paragraph 4.2.4.2: Add at end of paragraph:

"When more than one unit is installed in the ship and when different units are oriented in the ship so that the principal axis of one unit is fore and aft and the principal axis of the other unit is athwartship, then the equipment shall be tested with each horizontal axis inclined on fixture, FIGURE 10-1 or 10-3 or equivalent. In this case, a minimum of nine blows is required. Equipment tested on the 30 degrees bulkhead fixture, FIGURE 10-2, will be qualified for either orientation in the ship and the six blow minimum is therefore applicable. Equipments whose orientation in the ship does not vary shall be inclined in the same direction as the equipment would incline were the ship to

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roll (roll is the motion of the ship around its fore and aft axis). The six blow minimum applies. This inclination of the equipment is to ensure that equipment will be tested in the same direction as if the ship were subjected to a shock from an underwater explosion from the athwartship direction. To facilitate installing some equipments on the shock machine, the equipment may be rotated 90 degrees in lieu of 30 degrees. Equipment which is installed at 90 degrees inclination is subject to the six blow minimum regardless of orientation in the ship. Where an equipment has more than one significant operating mode, the series of blows shall be programmed to test the equipment in each of these modes. In this case, the mode of operating and the number and type of blows for each mode shall be specified. The equipment shall undergo a Group III blow in each orientation in each significant mode, except that, regardless of the number of modes, the total number of blows shall not exceed 18. A separate item of equipment may be submitted for each series or part of a series of six blows in excess of six blows, if desired by the manufacturer."

Page 10, TABLE I: Delete: "250" and substitute "0".

Page 10, paragraph 4.2.4.3: Delete and substitute:

"4.2.4.3 For heavyweight test (internal equipment). Heavyweight test for internal equipment shall be conducted on the floating shock platform shown on FIGURE 3. The test series shall consist of five shots using standard 60 pound test charges suspended 24 feet blow the water surface, at horizontal ranges of 60 feet (ft), 40 ft, 30 ft, 25 ft, and 20 ft, from the near side of the platform. Shot 2 (40 foot standoff) shall be oriented with respect to the platform so as to load the equipment in the same manner as would an explosion occurring off the bow or stern of the ship in which the equipment is to be installed. All other shots will simulate explosions from an athwartship direction. Except for shot 2, failures experienced during any particular shot shall be corrected and proven on the next shot, for example, an equipment which experienced a failure on shot 4 (25 foot standoff) may be accepted after corrective action and upon satisfactorily withstanding shot 5 (20 foot standoff). A failure experienced during shot 5 will necessitate a sixth shot (repeat of 20 foot standoff), unless the failure is considered by the acceptance authority to be clearly capable of correction by post-test design modifications (see 4.2.7.2(a)(3)). A failure experienced during shot 2 will necessitate a repeat of shot 2. NAVSHIPS Instruction 9690.7 lists those activities which operate floating shock platforms. The commercial activities may be contacted directly regarding the use of these facilities. NAVELEX should be contacted regarding the use of the Government facilities (see 4.2.9). In certain cases, for subsidiary components or subassemblies, a modified test schedule may be specified based upon the results of shock analysis performed on the associated principal unit."

Page 10: Add new paragraph 4.2.4.4:

"4.2.4.4 For heavyweight tests (external equipment). Tests for external hull-mounted equipment shall be conducted on the floating shock platform shown on FIGURE 3. Unless otherwise specified in the individual equipment specification, the test series shall consist of five shots using 60 pound charges suspended 24 feet below the water surface, at horizontal range of 60 ft, 40 ft, 30 ft, 25 ft, and 20 ft from the near side of the platform. Requirements for equipment operation, orientation, inspection, instrumentation, and similar facilities shall be specified by the Bureau or Agency concerned."

Page 11: Add new paragraph 4.2.5.3.1:

"4.2.5.3.1 The shock test report shall include information on the item's hold-down bolts that were used in the shock test. The information shall include the material, size, and type of the hold-down bolts. Also, where prestressing of fasteners is required, the torques on the fasteners will be recorded before and after the shock test, and where prestressing is not required loosening of the fasteners during the shock test will be recorded in the shock test report. Shock test report shall also state type, model and number of resilient mounts in installation."

Page 11: Add new paragraphs 4.2.5.5 and 4.2.5.6:

"4.2.5.5 Where two or more items are installed on a common subbase and the total operating weight is less than 60,000 pounds, the assembly shall be shock tested. Equivalent dummy weights may be used to simulate all but one of a group of identical units installed on a common subbase.

"4.2.5.6 Post-shock test functional testing and inspection. The contractor or vendor shall prepare such requirements for the item and shall submit the requirements, including associated acceptance criteria, to the review and acceptance authority (see 4.2.7.2) for approval as part of the test report. This approval shall be obtained prior to shock testing for heavyweight equipment. The following minimum requirements shall apply to the post-shock test performance testing and inspection specifications:

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(a) Functional testing - After shock testing and unless otherwise specified, prior to disassembly or repair, and prior to the installation in the ship, the equipment shall be given suitable performance and other tests as required to determine whether or not it meets the specified operational requirements. In general, the tests shall include, but not necessarily be limited to, checking the input-output of the component or equipment, its operating temperatures (bearing, coil windings), and cyclic tests, as appropriate. Hydraulic, pneumatic, and fluid systems equipment shall be hydrostatically tested in accordance with the applicable component specification to demonstrate strength and to test for leaks, if not previously checked during shock testing. Electrical equipment shall be tested for shorts to detect breakdown of insulation. Grade A equipment shall be considered to have failed the shock test if the equipment is unable to meet the specified operational requirements. Grade B equipment need only demonstrate that no portion of the equipment will come adrift or otherwise create a hazard to personnel or Grade A systems and equipment. Failure of a Grade B unit to perform its principal function after shock shall not be cause for rejection.

(b) Inspection - In general, all shock-tested equipment shall be disassembled and inspected for breakage, deformation, and misalignment. Areas highly stressed during shock response and areas suspected of yielding shall be dye penetrant or magnetic particle checked for cracks. Critical tolerance areas shall be checked to ensure proper clearances. Equipment shall be considered to have failed the shock test if inspection reveals any condition that could prevent the equipment from performing its intended function, or could create a personnel hazard as a result of the shock. Small cracks, yielding of structure, out-of-tolerance clearances and similar damage will not be cause for disapproval of the equipment, unless such damage causes impairment of equipment function or substantially shortened equipment life."

Page 12, paragraph 4.2.7.2: Delete in its entirety and substitute:

"4.2.7.2 Test report. A test report for all shock-tested equipment shall be submitted to the Supervisor or other designated approval authority for approval (See FIGURES 14 through 18). Copies of each report shall be furnished to NAVELEX, Washington, D.C. 20363, for information. Copies of each report shall be furnished to the Naval Underwater Systems Center (NUSC), Code 342, Newport, RI 02840 for information and data storage purposes. The test report shall include the following:

(a) For lightweight and medium weight tests:

(1) A report by the shock-test facility (including FIGURES 14 and 15 or 16, as applicable) and clear photographs of each test setup and each instance of damage (see 4.2.5.3).

(2) A report by the activity (manufacturer or other) performing the post-shock test inspection and functional testing of the equipment (including FIGURE 18 (see 4.2.5.4)).

(3) A description of the corrective measures taken and recommendations as to proposed design changes which will correct any deficiencies found during the post-test inspection. (In certain cases, the Government may accept the equipment as shock qualified on the basis of the corrective design changes rather than to require retesting.)

(4) Certification of the report describing the shock test shall be by the facility performing the shock test.

(5) Certification of the post shock test inspection and functional testing report shall be by a Government representative. The Government representative need not witness all tests, but it is desired that a sampling of the tests by each test facility be witnessed by the Government representative.

(b) For heavyweight tests:

(1) As in (a)(1), above, except that information in FIGURE 17 is applicable.

(2) As in (a)(2), above.

(3) As in (a)(3), above.

(4) Certification of the report describing the shock test shall be by a Government representative who shall actually witness the test.

(5) As in (a)(5), above, except that the Government representative must witness the actual inspection and functional tests.

(c) All lightweight and medium weight test and post-test inspection reports for equipment shall be reviewed by and all such equipment shall be accepted or rejected for the Navy by the cognizant Supervisor of Shipbuilding, U.S.N., or Naval Shipyard or other Navy procuring agency (for example, NAVELEX). The supervisor or Naval Shipyard or other Navy activity may delegate the review and acceptance authority to other Government activities or representatives. If this is done, the procuring activity shall designate the review and acceptance authority in its procurement documents. All heavyweight test and post test inspection reports shall be reviewed by and all such equipment shall be accepted or rejected by NAVELEX or other designated approval authority."

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Page 12, paragraph 4.2.7.3: Delete in its entirety.

Page 12, paragraph 4.2.8: Delete "NAVSHIPS 250-423-30" and "NAVSHIPS 250-423-31"; add, "SUPSHIP THREE Report 280-8", "Design Data Sheet DDS 072-1" and "Naval Research Laboratory Report 7366".

Page 12: Add new paragraph 4.2.9:

"4.2.9 Use of Government shock facilities. Unless otherwise specified, shock tests shall be conducted at commercial test facilities. Requests for use of Government-owned shock test facilities will be honored, as scheduling permits, when accompanied by certification that commercial test facilities are not available. The following information must be furnished to NAVELEX with the request for use of Government-owned shock test facilities. (This same information shall be forwarded to the designated Government test facility at least two months prior to the scheduled test):

(a) Three copies of Equipment Identification, FIGURE 14.
(b) Drawings or technical manuals showing:
(1) Outline of equipment.
(2) Sectional assembly of equipment.
(3) Foundation for equipment.
(4) Bedplates and mounting when applicable.
(5) Arrangement in ship.
(6) Plans showing all connections to the equipment, such as piping, electrical conduct cables, air ducting, etc., which may affect the dynamic response of the equipment. These plans shall show the methods of attachment of these connections to the equipment and to the ship's structure, and their arrangement with respect to the equipment.
(7) Weight and center of gravity of the equipment and test foundations and fixtures.
(c) Three copies of the equipment purchase specification.
(d) Agenda and requirements for all equipment inspections to be performed at shock facility.
(e) Requirements for shock test instrumentation, if any. (Use of shock test instrumentation is not required by these specifications.)
(f) Any other special requirements or precautions associated with the shock test.
(g) Outline of required test configuration, detailed design of test fixture, and test conditions as outlined in 6.1 of MIL-S-901 (3 copies). This data will be subject to review and approval by NAVELEX to fulfill requirements of this section."

Page 16, paragraph 6.2.2: Delete in its entirety.

Page 16, paragraph 6.3 (see Amendment 1 of 5 Sept 1963) Delete and substitute:

"6.3 Test report. For shock tested equipment, the applicable test report forms are shown in FIGURES 14 through 18 of this specification. Until the supply is exhausted, the use of NAVEXOS 3373 forms may be continued for reports of lightweight and medium weight tests. Pads of NAVEXOS 3373 forms may be obtained upon application to the cognizant Defense Contract Administration Services (DCAS) office, except that activities of the Department of Defense should make application to the Navy Supply Center, Norfolk, VA 23412 and the Naval Supply Center, Oakland, CA 94625."

After page 31, add FIGURES 10-3 and 13 through 18 as follows:

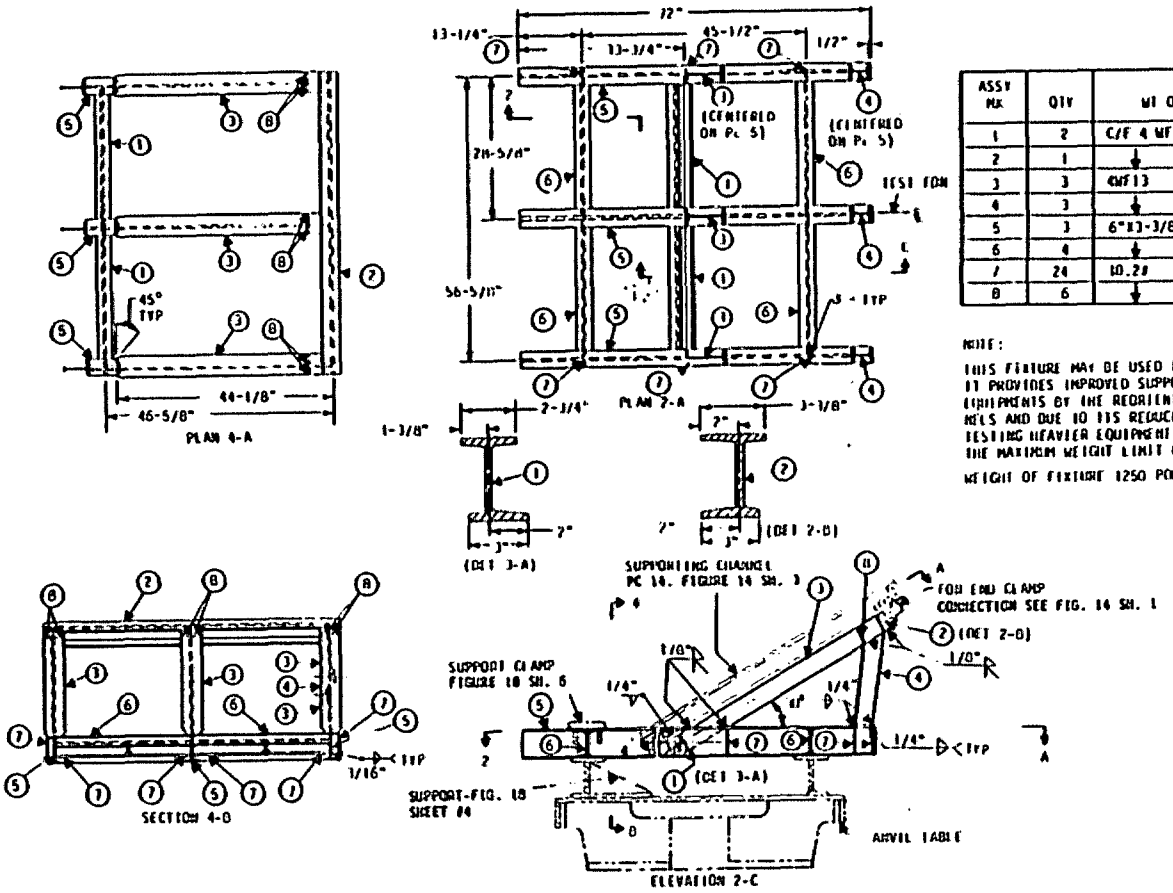
30.2 Underwater explosion qualification of submarine hull penetrations. In addition to the mechanical shock test requirements of MIL-S-901 relating to the continued mission capability of the submarine, the following criteria for maintaining watertight integrity during and after a near lethal attack by underwater explosion are established. Loadings resulting from the hull penetration underwater explosion testing are much more severe than those specified above for mechanical shock. Items which have been or will be qualified by the hull penetration underwater explosion test are exempt from the tests specified in MIL-S-901, if after underwater explosion test the items meet the acceptance criteria of MIL-S-901.

30.3 Design criteria. Components specified below shall be capable of withstanding the hull penetration underwater explosion test.

(a) Those items which penetrate the pressure hull, including but not limited to, hull and backup valves, diesel exhaust backup and hull valves, induction backup and hull valves, access and loading hatches, logistic/escape trunks, main shaft seal housing assembly including the stuffing box, launcher assembly, and trash ejector assembly

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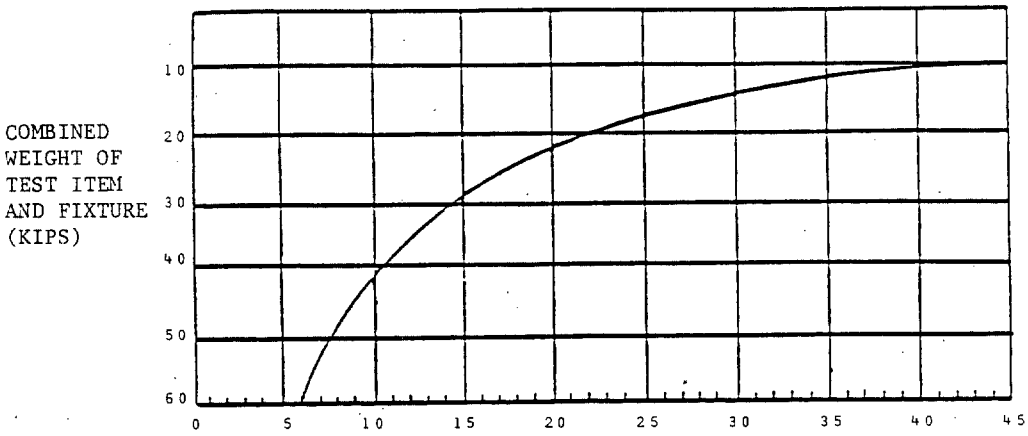
ASSY NO.	QTY	WT OR SIZE	GR/CL
1	2	C/F 4 WF 13	HT-TY A
2	1	↓	↓
3	3	4WF13	
4	3	↓	
5	3	6"X3-3/8"X12.5#	
6	4	↓	↓
7	24	10.2#	HT-TY 1
8	6	↓	HT-TY 1

FIGURE 10-3. 30 degree mounting fixture for testing base mounted equipment on medium weight shock testing machine

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APPENDIX

FLOATING SHOCK PLATFORM
MAXIMUM HEIGHT OF VERTICAL CENTER OF GRAVITY
FOR TESTED ITEMS
14' x 25' WORKING AREA - MAXIMUM
16' x 28' SIZE OVERALL



NOTES:
1. This information is provided for guidance only.
2. These maximums may be exceeded at the discretion ofthe Shipbuilder and shock testing facility.
3. This is for the platform without an outrigger. It is based on a transverse metacentric height of slightly more than one foot.

FIGURE 13. Floating shock platform maximum height of vertical center of gravity for tested items
14' x 26' working area -- maximum
16' x 28' size overall

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Page 1 of _____ Test No _____ Date _____

Class III Shock Test Report
Equipment Identification

1. Item (Name/Type/Nomenclature/Rating)
2. Manufacturer (Name & Address)
3. Model No. _____ 4. Size/Capacity _____
5. Service/Ship _____ 6. Technical Manual No. _____
7. Plan Numbers: Sectional Assy _____ Outline _____
8. Approximate Overall Envelope Size: _____ Long _____ High _____
9. Weight: Wet _____ Dry _____
10. Height of Center-of-Gravity Above Base of Equipment: _____
11. Contract or Purchase Order No. _____
12. Requirements (Para 3 MIL-S-901) (NAVY) Grade _____
    Classification, _____ Class _____
13. Test Procedure Identification _____
14. Major Components and Attached Items Included in Text

	Name	Identification	Manufacturer
a.			
b.			
c.			
d.			
e.			

15. Disposition of Equipment Following Test

16. Remarks

17. Distribution of Report

18. Test Laboratory
19. Address

FIGURE 14. Class HI shock test report.

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Page_____of_____Test No._____Date_____

Lightweight Shock Machine Test

(Attach Separate Sheet Wherever Necessary to Fully Describe Item)

1. Type of Test Fixture (Circle one)

   4A 4C 6D-1 6D-2 6E Other

   (attach photo(s) showing test set up on machine)

2. Total Weight on Anvil Plate (Equipment plus fixture)_____lbs.

3. Instrumentation, if any.

   Type Gauge Location Orientation Results

4. Monitored Performance

Blow No.	Drop Feet	Axis	Operating Mode (on, off, open, close etc)	Reference Measurements (volts amps rpm psi etc) a.	b.	c.	d.	Post-Test Measurements volts amps rpm psi etc a.	b.	c.		
1	1	back										
2	3	back										
3	5	back										
4	1	side										
5	3	side										
6	5	side										
7	1	top										
8	3	top										
9	5	top										

5. Survey Findings (Note damage, if any, and attach photos, if available)

6. Modifications, if any, accomplished prior to or during test

7. Remarks

CERTIFICATION OF TEST REPORT

Test Laboratory	Address	Signature of Test Engineer

FIGURE 15. Lightweight shock machine test.

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Page _____ of _____ Test No. _____ Date _____

Medium Weight Shock Machine Test

(attach Separate Sheet Whenever Necessary to Fully Describe Item)

1. Type of Test Fixture (Circle one)

   Fig 9 Fig 10-1 Fig 10-2 Fig 13 Other

   (attach photo(s) showing test set up or machine)

2. Total Weight on Anvil Table (Equipment Plus Fixture) _____ lbs.

3. Instrumentation, if any.

   Type Gauge Location Orientation Results

4. Monitored Performance

Blow No.	Group No.	Drop Feet	Operating Mode (on, off, open close etc)	Reference Measurements volts amps rpm psi etc				Post-Test Measurements volts amps rpm psi etc			
				a.	b.	c.	d.	a.	b.	c.	d.
1	I										
2	inclined										
3	II										
4	inclined										
5	III										
6	inclined										

5. Survey Findings (Note damage, if any, and attach photos, if available)

6. Modifications, if any, accomplished prior to or during test.

7. Remarks

CERTIFICATION OF TEST REPORT

Test Laboratory	Address	Signature of Test Engineer

FIGURE 16. Medium weight shock machine test.

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Page_____of_____Test No._____Date_____

Heavyweight Shock Test on Floating Shock Platform (FSP)

This sheet outlines required information only. Additional pages will undoubtedly be required.

1. Type of test fixture and details of installation
2. Total weight installed on Floating Shock Platform (FSP)_____
3. Instrumentation (include sketch or photo if necessary)

   Type Gauge Location Orientation Results

4. Monitored Performance

Shot	Range	Date	Operating Mode (on, off running, idle rated speed, etc)	Reference Measurements: volts amps rpm psi etc				Post-Test Measurements volts amps rpm psi etc			
				a	b	c	d	a	b	c	d
1	60										
2	40										
3	30										
4	25										
5	20										
6											

5. Survey Findings (Note damage, if any, and include photos)
6. Modification, if any, accomplished prior to or during test.
7. Remarks

Government Activity	Address	Signature of Government Rep.

FIGURE 17. Heavyweight shock test on floating shock platform (FSP).

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Page_____of_____Date_____

Refer to Test Lab_____Test No._____of Date_____

Post Shock Inspection Report

TO:

FROM:

Contract (s) -

Component Tested -

Manufacturer -

Component Dwg. No. -

Contractor Purchase Order -

Type of Test (check one)	Specification	Test Report No.
HI Shock Test	MIL-S-901	
Hydraulic Shock Test		
Sub-Hull Penetration Test		
Other		

Inspection Method	Check as Applicable		Inspection Initial and Date			
	Required	Accomplished	Vendor	Date	Gov't	Date
1. Hydrostatic Test						
2. Dye Penetrant Insp.						
3. Magnetic Particle Insp.						
4. Radiographic Insp.						
5. Dielectric Test						
6. Electrical Continuity						
7.						

Repairs (were) (were not) necessary. If necessary, describe fully under remarks attach separate sheet(s).

REMARKS:

Disposition of Unit_____

I certify that the above represents a true report of the tested unit.

Government Inspector

FIGURE 18. Post shock inspection report.

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(b) Those items in piping systems subject to sea pressure internally and which are between the sea and the inboard flanged joint of the backup valve

(c) Periscope hull fittings

(d) Each size of each design of fittings must be tested for qualification, except that for sizes eight inches and smaller, testing to two sizes of a design no more than four inches apart will qualify all sizes in between. For allowed size spread for valves, see MIL-STD-798, Section 11

(e) Those items, such as external hatch covers, which could block passage through logistics/escape trunks

(f) Items previously qualified by underwater explosion testing need not be requalified. Items which are similar in design, construction, installation, and function to items which have previously satisfied the hull penetration test requirements of this specification may be considered acceptable for this submarine with no further testing required, subject to NAVELEX approval. Criteria and requirements applicable to extension of hull penetration test qualification are equivalent (as applicable) to those described in MIL-S-901 for shock test qualification extension.

30.4 Contractor responsibilities. The following shall be provided to NAVELEX at least four months prior to the scheduled shock test date to support the underwater explosion test:

(a) Test vehicle installation drawings and procedures, including detailed hanger arrangement and type to simulate the ship installation, bolt torques, and method of simulating piping system loading. NAVELEX approval of the overall installation drawings is required.

(b) A test procedure which details the data which should be obtained, including any special instructions required such as valve assembly and disassembly procedures and bolt tightening procedures. NAVELEX approval of the test procedure is required.

30.4.1 Test items. The contract shall provide the test item together with any necessary prefabrication work to permit installation on the test vehicle at least one month in advance of the scheduled test date. Hull penetrations furnished for test shall be complete and ready to install and shall include the following:

(a) Operators and handwheels for valves

(b) Piping runs up to the second hanger or to the anchor point, with necessary hangers

(c) Hold-down bolts required for installation

(d) Masses to simulate weight and moment of missing components of a test assembly, such as a dummy capstan head or loads due to connecting systems (excepting piping)

(e) Hull inserts required for support of valves and fittings scheduled for test

The contractor shall provide a test site representative to facilitate timely testing and evaluation of items. Valves, fittings, and inserts used for the test shall not be installed on the ship.

30.4.2 Hull penetration test procedure.

30.4.3 General. The underwater explosion test will be conducted using the Full Scale Section Number 5 as the test vehicle. Items subjected to the underwater explosion test shall be structurally supported in a manner representing the most vulnerable shipboard installation and shall include weights as necessary on the inboard connections to simulate the loads imposed by connecting systems. When testing items in the closed position, the cavity inboard of the outboard closure shall be filled with the fluid normally used in service (for example, in testing sea water valves use water). For items which are actuated by a hydraulic operator, the hydraulic operator, shall be pressurized to the maximum system operating pressure normally required to move the operator.

30.4.4 Double closure test procedure. Piping runs between inboard and outboard closure shall be simulated in the test installation. The test series for components having inboard and outboard closures shall consist of three shots, each of 125 lbs. of HBX-1, with components on a horizontal radius from the centerline of the test vehicle and the charge on an extension of that radius at a standoff of 15 ft (from the closest pressure boundary of the component or test vehicle) for the first two shots and 13.5 ft for the third shot. For the first shot, the outboard closure shall be shut and the inboard closure open. For the second shot this shall be reversed. For the third shot the components shall be in the normal operating position.

30.4.5 Single closure test procedure. The test series for components having only a single closure (that is, no inboard closure) shall consist of two shots, each of 125 lbs. of HBX-1, with the components on a horizontal radius from the centerline of the test vehicle and the charge on an extension of that radius at a standoff of 15 ft (from the closest pressure boundary of the component or test vehicle) for the first shot and 13.5 ft for the second shot. For the first shot the closure shall be in the alternate position (that is, opposite to normal position) and for the second shot the closure shall be in the normal operating position.

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30.4.6 Fittings, packings, and seals test procedure. The test for electrical, electronics, or mechanical fittings, packings, or seals (except main shaft seal) penetrating the pressure hull and having only one operating position shall consist of two shots of 125 lbs. of HBX-1, with the fitting or component on a horizontal radius from the centerline of the test vehicle and the charge on an extension of that radius at a standoff of 15 ft (from the closest pressure boundary of the component or test vehicle) for the first shot and 13.5 ft for the second shot.

30.4.6.1 Special fittings or components. Special fittings or components may require test procedures which differ from those given above. The detailed special procedures where not specified herein, shall be as agreed upon between the contractor and NAVELEX.

30.4.7 General inspection and acceptance criteria. Tests shall be conducted at the Government test facility designated by NAVELEX. After each shot, components shall be visually inspected, without dismantling, for leakage and/or distortion of the watertight envelop, as defined in the test procedure. Those components with moving parts such as valves and hydraulic operators, shall be cycled twice to prove operability. Torques and/or pressures required to cycle components shall be recorded. Following the completion of operability tests, closures shall be hydrostatically tested to the submergence pressure specified. (The pressure shall be applied at the sea side of the closures.)

30.4.7.1 Bolts. Following the completion of hydrostatic tests, bolts shall be checked for tightness and retightened, if necessary, to the specified torque. The amount of retightening required and the calculated loosening thus indicated shall be recorded.

30.4.7.2 Distortion. Distortion will be permitted provided the distortion does not degrade the watertight integrity or operational capabilities.

30.4.7.3 Damage or leakage report. After each shot, if damage or leakage is within the acceptable limits specified herein, the componets shall be considered ready for the next shot. If, after any shot, damage is of a more serious nature (such as, loss of watertightness or operability), the component shall be disassembled by the test facility as necessary to evaluate and report extent of damage to NAVELEX, the Design Agent and Shipbuilder. The test facility shall include the following in its evaluation:

(a) Type and extent of damage
(b) Ability of Shipbuilder to make repairs at the test site
(c) Schedule and cost information as appropriate

30.4.7.4 Acceptance criteria. After the final shot the component and watertight envelope shall be given a final examination for leakage, operability, and damage as defined in 30.4.7.3 and in the test procedures. Also after the final shot, nonpiping system components shall be disassembled and inspected for evidence of mechanical binding, distortion or changes in electrical characteristics, and shall have their housings subjected to magnetic particle or dye penetrant inspection as appropriate. Watertightness and operability acceptance criteria for items are specified below:

(a) Double closure acceptance criteria. For components having inboard and outboard closures, the following acceptance criteria shall apply:

(1) After shots 1 and 2, the outboard closure and the inboard closure shall be operable using the normal operating procedures through two full cycles without exceeding the full design torque of the actuator. For valves so designed, the pressure may be equalized within the valve envelope prior to the operability test. In addition, the envelop (consisting of the connection of the insert to the pressure hull inboard to and including the outboard closure seal or seat, and also inboard to and including the inboard closure seal or seat) shall remain watertight when subjected to hydrostatic pressure as specified in 30.4.7, except for valves, where leakage past each valve seat shall not exceed 0.7 gal/min/inch of valve port diameter.

(2) After shot 3, at least one closure (either the inboard or outboard) shall be operable using the normal operating procedure through two full cycles without exceeding the full design torque of the actuator. For valves so designed, the pressure may be equalized within the valve envelope prior to the operability test. In addition, the envelope (consisting of the connection of the insert to the pressure hull inboard to and including the closure seal or seat of the operable closure) shall remain watertight when subjected to the hydrostatic pressure as specified in (1) above, except for acceptable valve seat leakage as specified in (1) above.

(b) Single closure acceptance criteria. Components having only a single closure must remain operable (using the normal operating procedure) and maintain a watertight envelope between the seat and seal of the closure after both shots.

(c) Fittings, packings, and sales acceptance criteria. Electrical and electronic fittings penetrating the pressure hull must remain operational and watertight following the 15-foot shot, and watertight following the final shot. Acceptance criteria for seals and packings shall be the same as for electrical and electronic fittings, except that seepage not to exceed the limits allowed for normal operation is permitted after any shot.

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(d) Access covers and closure plate acceptance criteria. Unless otherwise approved by NAVELEX, leakage across covers and closure (regardless of size) shall not exceed 0.5 gal/min after any shot, with leakage measured prior to retightening the cover or closure plate bolting. Furthermore, it must be demonstrated that this leakage can be reduced to an amount not to exceed 0.05 gal/min by retightening the cover or closure plate bolting.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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